

Letter

Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

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Authors

Mr. Lucas Buffan
Corresponding Author
Submitting Author

 [ORCID](https://orcid.org/0000-0002-2353-1432)
<https://orcid.org/0000-0002-2353-1432>

Affiliations
• Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France

Dr. Fabien L. Condamine

 [ORCID](https://orcid.org/0000-0003-1673-9910)
<https://orcid.org/0000-0003-1673-9910>

Affiliations
• Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France

Dr. Philip B. Holden

 [ORCID](https://orcid.org/0000-0002-2369-0062)
<https://orcid.org/0000-0002-2369-0062>

Affiliations
• Environment, Earth and Ecosystem Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK

Dr. Sara Varela

Affiliations
• Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

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Buffan_etal-GABI_Main_text.docx	Main Document - MS Word	2.6 MB	Page 3
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1 Palaeoclimate alone fails to explain the asymmetry of the Great 2 American Biotic Interchange

3 Lucas Buffan^{1*}, Fabien L. Condamine¹, Philip B. Holden², Sara Varela³

4 ¹ Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place
5 Eugène Bataillon, 34095 Montpellier cedex 5, France

6 ² Environment, Earth and Ecosystem Sciences, The Open University, Walton Hall, Milton Keynes,
7 MK7 6AA, UK

8 ³ Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de
9 Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

10 * Corresponding author: lucas.l.buffan@gmail.com

11 Statement of authorship

12 S.V. and L.B. designed the project. P.B.H. gave inputs on palaeoclimate data usage. L.B. ran the
13 simulations, performed the analyses, produced the figures and wrote the manuscript with inputs from all
14 co-authors.

15 Data availability statement

16 R scripts for generating, analysing and visualising the data are available at
17 <https://github.com/Buffer3369/GABI-gen3sis.git>. Simulation landscapes and config files are available at
18 <https://doi.org/10.6084/m9.figshare.31160704>.

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1 Abstract

2 Asymmetric colonisation patterns often arise from large-scale biotic exchanges involving two
3 source areas. In the Pliocene, the closure of the isthmus of Panama facilitated the admixture between
4 North and South American faunas, an event termed Great American Biotic Interchange (GABI). The
5 ecological mechanisms underlying the asymmetry of the GABI, incontestably in favour of North
6 American lineages, remain debated. Here, coupling a mechanistic eco-evolutionary simulator with
7 palaeoclimatic reconstructions, we test fundamental hypotheses relating this biogeographic imbalance to
8 palaeoclimate dynamics. When climate is the only constraint to species' dispersal, our simulations yield
9 more South-to-North than North-to-South dispersal events, mostly within the tropical biome. Also, our
10 models fail to explain the migrant diversity disequilibrium between the two continents, as well as
11 temperate specialists' over-representation in the fossil record of successfully-exchanged taxa. Hence, we
12 argue that spatio-temporal palaeoclimate dynamics alone cannot explain the asymmetry of the GABI,
13 suggesting additional factors fashioned this macroecological enigma.

14

1 Introduction

2 The Great American Biotic Interchange (GABI) refers to the large-scale biotic exchange that
 3 occurred between the Americas and intensified after the closure of the isthmus of Panama (Marshall *et al.*
 4 1982; Stehli & Webb 1985; Webb 1976). Though the exact timing of the appearance of such a landbridge
 5 is not entirely resolved (Bourke *et al.* 2025; Jaramillo 2018; Montes *et al.* 2015; O’Dea *et al.* 2016),
 6 palaeontological evidence and molecular phylogenies both agree with a late Miocene onset of (non-
 7 volant) biotic exchange between the Americas (Carrillo *et al.* 2015; Tedford *et al.* 2004; Verzi *et al.*
 8 2008), with effective exchange pulses occurring during the Plio-Pleistocene (Bacon *et al.* 2015, 2016;
 9 Marshall *et al.* 1982; Woodburne 2010). Among land mammals, a puzzling feature of the GABI is that it
 10 impacted the North and South continental faunas very differently. Although evidence from the fossil
 11 record indicates that the early phases of the GABI were marked by reciprocal land mammal exchanges
 12 between the Americas (Marshall *et al.* 1982; Webb 1976; Woodburne 2010), during the Pleistocene, the
 13 ability of migrant lineages to maintain and diversify in the newly-colonised continent became greater in
 14 South America (Woodburne 2010). Many native South American lineages concomitantly declined, some
 15 of which even going totally extinct (Boscaini *et al.* 2025; Carrillo *et al.* 2020; Cione *et al.* 2015; Croft *et*
 16 *al.* 2020; Webb 1991). Hence, this intercontinental faunal exchange was asymmetric.

17 Although this macroecological imbalance is unquestionable, no clear consensus regarding the
 18 drivers underlying this asymmetry has yet been established. On the one hand, biotic interactions involving
 19 migrant and native species have often been proposed to explain such an asymmetry. These would include
 20 competition (Domingo *et al.* 2020; Simpson 1980; Webb 1976) or predation of predator-naïve preys
 21 (Faurby & Svenning 2016; Fernández-Monescillo *et al.* 2025; Marshall & Cifelli 1990; Webb 1976).
 22 However, their contribution to the asymmetry of the GABI has been increasingly called into question
 23 (*e.g.*, Freitas-Oliveira *et al.* 2024a; Marsà *et al.* 2025; Prevosti *et al.* 2013; Tarquini *et al.* 2022; Vrba
 24 1992). On the other hand, several studies proposed that the asymmetry of the GABI emerged from
 25 palaeoclimate dynamics (*e.g.*, Bacon *et al.* 2016; Freitas-Oliveira *et al.* 2024b; 2025; Marshall & Cifelli
 26 1990; Vrba 1992; Webb 1991). The prevailing views consist of relating pulses of transcontinental
 27 mammal exchanges with Pleistocene climatic oscillations, promoting the transition from tropical
 28 rainforest to savannah-like environments in Central America as climate cooled down (Figure 1). Based on
 29 consideration of the migrant fossil record across both sides of the isthmus of Panama, Webb (1991)
 30 proposed that this climate-driven landscape opening would have facilitated the dispersal of open-adapted
 31 taxa. As those were assumed to be more numerous in North than in South America – the former continent
 32 offering more high-latitude temperate-to-cold environments than the latter –, a larger number of North
 33 American taxa would be predicted to disperse to the South than the other way around (Figure 1 A). The
 34 resulting exchanges would have been periodically interrupted by intervals of warming, promoting the

reverse transition to rainforest in the isthmian region, with rainforests acting as a barrier for species dispersal (tropical ‘holding pen’ hypothesis) (Woodburne 2010). In addition, Vrba (1992) proposed the contraction of tropical habitats towards the equator promoted by the aforementioned periodical climatic cooling to have mechanically attracted taxa suited to tropical environments southwards (Figure 1 B). However, studies supporting these views have only been descriptive, tackling the issue from a pattern-oriented perspective driven by fossil data but, to our knowledge, never from a process-oriented perspective.

Recent improvements in process-explicit modelling have marked a turning point in the study of biodiversity (Pilowsky *et al.* 2022). By combining a landscape including palaeoenvironmental variables and a set of eco-evolutionary rules according to which virtual species are simulated, mechanistic eco-evolutionary models offer the unique opportunity to causally explore how biodiversity patterns arise and are maintained through evolutionary times (Hagen 2023). Recently, this class of models has provided unprecedented insights into the underlying forces behind widely-recognised biodiversity patterns, such as the latitudinal biodiversity gradient (Hagen *et al.* 2021a; Lorcery *et al.* 2025), the diversity disequilibrium between tropical rainforests across the globe (Hagen *et al.* 2021b), the build-up of the unique biodiversity along the Wallace line (Skeels *et al.* 2023b) and many others (*e.g.*, Boschman *et al.* 2023; Liu *et al.* 2024; Skeels *et al.* 2023a).

Here, we test Webb and Vrba’s hypotheses on the role of palaeoclimate dynamics in the asymmetry of the GABI (Figure 1) using a spatially-explicit mechanistic biodiversity simulator coupled with palaeoclimate reconstructions across the last 5 million years. Our simulations tracked the diversification of a clade originating either in North or South America. The ability of a virtual species to survive at a given place was only mediated by climatic factors. We hypothesise that (i) simulations starting from a North American ancestor would be more successful at reaching the other continent than the other way around, (ii) successful simulations starting from a North American ancestor would result in a more diverse and spatially spread radiation in the newly-colonised continent, (iii) niche evolvability and (iv) initial biome would significantly impact exchange frequencies, with higher values obtained for adaptive-niche simulations, or fixed-niche simulations with temperate or arid rather than tropical affinity.

1 Material and Methods

2 Beforehand, it has to be noticed that, across the entire study, ‘North America’ actually refers to the
3 current North and Central American continents. The delimitation between North and South America was
4 arbitrarily set as a straight line connecting two points located along the eastern and western coasts of
5 present Panama (82.882° W, 7.138° N and 81.369° W and 9.008° N) (see Figure S1).

6 Palaeoclimate data

7 We used the spatio-temporal terrestrial palaeoclimate series from the PALEO-PGEM emulator
8 (Holden *et al.* 2019) of the PLASIM-GENIE intermediate-complexity atmosphere-ocean general
9 circulation model (Holden *et al.* 2016). This reconstruction covers the last 5 Million years (My)
10 downscaled to a $1 \times 1^\circ$ spatial resolution (Barreto *et al.* 2023). For the scope of this work, we used two
11 bioclimatic variables: mean annual temperature (MAT, BIO1) and mean annual precipitation (MAP,
12 BIO12). To avoid an intractably-high number of simulation time steps, we reduced the temporal
13 resolution of the climate simulations from a 1000-years to a 10,000-years time step by sampling
14 corresponding time slices. This procedure did not alter the overall climatic trends. In particular,
15 Pleistocene climatic oscillations clearly remained after downscaling (Figure S12).

16 In order to discriminate simulations based on the biome where the initial species originates, we
17 reclassified monthly palaeoclimate layers for the starting time ($t = 5$ My ago) according to Köppen-
18 Geiger broad climate types (Beck *et al.* 2018; Galván *et al.* 2023). This classification divides the world
19 into five broad climatic regions: Polar, Cold, Temperate, Arid and Tropical, following the criteria
20 proposed by Köppen (1900).

21 Simulation model and modelling schemes

22 We performed spatially-explicit simulations of biodiversity using gen3sis v1.5.11 (Hagen *et al.*
23 2021a). Simulations, moving forward-in-time with a 10ky time step, tracked the radiation of a clade
24 arising from a single ancestral population, either originating in North or in South America, throughout the
25 last 5 My. In gen3sis, virtual species are modelled with individual populations evolving independently,
26 following a set of four rules: dispersal, speciation, environmental filtering and trait evolution. All
27 simulation scenarios that we implemented only differed on the last rule (trait evolution), the first three
28 being common to all models.

29 We elaborated a gridded landscape from the time-series of the two palaeoclimatic layers. Within
30 this landscape, populations making up virtual species inhabited a set of connected cells. At each time step,
31 inter-cell dispersal could occur with an intensity sampled in a Weibull distribution (repeated
32 independently for each species-cell combination). As done by Skeels *et al.* (2023a); (2023b), the cost of

1 moving from a cell to another was made twice harder if the target cell was not habitable (*i.e.*, ocean or
 2 sea). If, by chance, two populations become disconnected, they accumulate divergence as long as they
 3 remain disconnected, at a rate ruled by a divergence factor, here set to 0.05 (Table S1). Once passed a
 4 divergence threshold, speciation would occur – note that, consequently, the only mode of speciation
 5 allowed by gen3sis is allopatry.

6 As our goal was to evaluate the extent to which palaeoclimate change may have shaped the unequal
 7 success of North American taxa establishing southwards compared to South American taxa establishing
 8 northwards, local climate was set as the only constraint for a population to survive in a given cell. Each
 9 species was therefore assigned a two-dimensional climatic niche, one axis corresponding to temperature
 10 suitability and the other to precipitation suitability. The abundance of a species along an environmental
 11 gradient was determined by two parameters – that we also call traits – : a population- and species-specific
 12 optimum (resp. $P_{k,l}$ and $T_{k,l}$ for the precipitation and temperature optima of the population k of the
 13 species l) and a breadth (or tolerance) (resp. ω_P and ω_T ; within a simulation, all populations and species
 14 had the same breadths). A population could survive in a given cell only if the difference between its
 15 optima and local climate (both temperature and precipitation) was less than its tolerance. Therefore, given
 16 a site i with local precipitation and temperature at time t being $P_i(t)$ and $T_i(t)$, this translates in

$$17 \quad \begin{cases} |P_i(t) - P_{k,l}| < \omega_P \\ |T_i(t) - T_{k,l}| < \omega_T \end{cases}$$

18 We implemented various simulation scenarios (also called modelling schemes) where we
 19 modulated the ability of species to cope with climate change. In the adaptive niche model ($M1$), climatic
 20 optima ($P_{k,l}$ and $T_{k,l}$) were able to evolve in the direction of local climate change. To isolate the effect of
 21 a single environmental covariate on the resulting simulated biodiversity pattern, alternative versions of
 22 this model consisted of either removing the temperature ($M2$) or the precipitation ($M3$) constraint on
 23 species ecology. Besides, we also implemented a fixed niche model ($M0$), where niche optima could not
 24 evolve and were therefore held constant (to their initial value) across all simulations. 500-member
 25 simulation ensembles were performed for each simulation scenario, and for both North and South
 26 America as ancestral area. For details about initialisation of a simulation and parameter values, please
 27 refer to *Supplementary text*.

28 Importantly, simulations generated under the fixed niche model ($M0$) originating in the tropical
 29 biome of either continent allowed to assess fundamental hypotheses related to the concept of tropical
 30 niche conservatism (Wiens & Donoghue 2004). This concept has often been invoked to explain the high
 31 diversity of tropical compared to temperate regions. In the framework of the Americas, Smith *et al.*
 32 (2012) evidenced that the extant vertebrate biodiversity across the Americas resulted from an asymmetric

1 niche conservatism. They indeed showed that taxa of North American origin were a significant compound
 2 of the high Neotropical biodiversity, whereas taxa with a southern ascent were little represented in
 3 temperate North America, therefore arguing that South American taxa exhibit an higher niche
 4 conservatism (in favour of tropical areas) than North American taxa. Thus, our modelling schemes also
 5 provide the ability to test whether palaeoclimate is a sufficient driver behind this observed niche
 6 conservatism asymmetry across the Americas.

7 Throughout the study period, a vast proportion the South American landmass was dominated by
 8 relatively stable tropical climatic conditions whereas the overall North American climate was dominated
 9 by temporal fluctuations (Figure S1). Consequently, we expect that lineages generated under fixed-niche
 10 simulations ($M0$, high conservatism) originating in South America to be less intensively exchanged than
 11 those generated under evolving-niche simulations starting in North America ($M1-3$, attenuated
 12 conservatism). Such an imbalance would be in line with the empirical pattern of exchange asymmetry that
 13 took place during the GABI.

14 Last, it is important to notice that, since the palaeoclimate reconstructions that we used ignored
 15 tectonic plate motion – a reasonable assumption for a 5-My time frame – continental position in our
 16 landscapes was held constant across simulations. Therefore, in that framework, changes in connectivity
 17 between North and South America could only arise from climate.

18 Quantification and statistical analyses

19 At every time step, for each run of each simulation, we used a function released by Skeels *et al.*
 20 (2023b) to save presence-absence matrices. Given a time t , such matrices stored on their columns the
 21 presence of each simulated species that have existed from the onset of the simulation to t within each cell
 22 of the landscape (rows of the matrix) as a binary variable – 1: the species inhabits that cell, 0: it does not.

23 Given a batch of simulations, after discarding those that crashed – *i.e.*, that led to no living
 24 representative – we quantified four metrics reflecting the occurrence, magnitude and efficiency of an
 25 eventual transcontinental exchange based on the presence-absence matrix of the last time step of each
 26 successful run (hereafter called t_{end}). First, we assessed the exchange success, a binary variable set to one
 27 if at least one pixel of the area to colonise (hereafter called A_{col} , *e.g.*, South America if the simulation
 28 started in North America) recorded a presence. Then, we recorded the biome in which the initial ancestor
 29 occurs (*i.e.*, Tropical, Arid, Temperate, Cold, Polar) as well as the distance of this initial ancestor to a
 30 point arbitrarily located on the zone of the isthmus of Panama, (-84°lon , 10°lat). Last, in the case of a
 31 successful exchange, we quantified the number of virtual taxa recorded in A_{col} at t_{end} – referred to as
 32 “colonised diversity” – and the proportion of A_{col} actually colonised at t_{end} as the ratio between the
 33 number of cells occupied by the colonisers and the total number of available cells (*i.e.*, surface of A_{col}).

1

1 Results

2 South-to-North dispersal events are made easier by climate

3 Our results suggest that the proportion of simulations successfully reaching the other continent is
 4 always higher – by approximately a factor of two – when starting from South America than from North
 5 America (Figure 2). This imbalance holds whether we allow species climatic niche to evolve following
 6 the direction of local climatic change ($44 \pm 6\%$ vs. $87 \pm 3\%$, model *M1*) or not ($29 \pm 6\%$ vs. $59 \pm 5\%$,
 7 *M0*), although we can notice that overall transcontinental exchange frequencies are lower in the latter
 8 case. When allowing climatic niches to evolve, amputation of either their temperature ($47 \pm 7\%$ vs. 88
 9 $\pm 3\%$, *M2*) or precipitation ($38 \pm 4\%$ vs. $89 \pm 3\%$, *M3*) compounds did not alter the pattern. We
 10 emphasise the fact that, contrary to our expectations, fixed-niche simulations originating in South
 11 America always yielded to a greater exchange intensity than evolving-niche simulations starting in North
 12 America. The simulated imbalance persisted after standardising initial distances to the isthmus of Panama
 13 for simulated species originating in North America to those of simulated species from South America,
 14 except for simulations generated under the fixed-niche model ($50 \pm 6\%$ vs. $59 \pm 5\%$) (Figure S4). Last,
 15 we stress that the simulated pattern was also robust to thermal breadth expansion for lineages originating
 16 in North America, up to 10 times broader than for simulations originating in South America (Figure S14
 17 A).

18 Out of the 500 runs carried out for each initial landmass-modelling scenario combination, the
 19 proportion of simulations leading to no living representative (=crash) was on average three times higher
 20 for simulations starting from North America, except for model *M3*, recording almost no crash (Table S2).
 21 This difference, although attenuated, persisted after the spatial standardisation procedure, but fully
 22 disappeared when we increased the thermal breadth of simulations originating in North America (Figure
 23 S14 B).

24 Successful establishment on the newly-colonised continent

25 Irrespective of the initial continent, our findings highlight that successful simulations generated
 26 under models allowing species climatic niches to evolve (*M1–M3*) colonised a larger area on the newly-
 27 colonised continent than those generated under the *M0* model – where climatic niche evolution was not
 28 allowed (Figure 3 A). Furthermore, species of North American origin that successfully made it to South
 29 America generally colonised a greater area than those of South American origin reaching North America
 30 (Figure 3 A). This pattern is particularly striking for simulations generated under the *M0* model ($36.5 \pm$
 31 3.5% vs. $11.7 \pm 0.4\%$), and holds for all but the *M3* model – relaxing the precipitation constraint on
 32 species' niche –, the latter suggesting an opposite trend ($69.9 \pm 3.3\%$ vs. $79.4 \pm 3.9\%$). However, due to
 33 the difference in size (*i.e.*, number of pixels) between the two continents (Figure S1), the actual colonised

1 area was greater when simulations generated under the *M1–3* models started from South than North
 2 America, but remained larger when simulations were generated under *M0* started from North America
 3 (Figure S11).

4 Besides, contrary to the proportion of colonised area for a given ancestral continent, colonised
 5 diversity does not vary significantly across our four models. However, a significant decrease is evident in
 6 simulations starting from North America generated under the *M3* model (Figure 3 B). Moreover,
 7 ancestral continent does not seem to affect mean colonised diversity values in the case of the *M1* ($212 \pm$
 8 61 vs. 170 ± 25 species respectively retrieved in the newly-colonised continent after starting from North
 9 or South America) and *M2* (161 ± 48 vs. 171 ± 27 species) models. It does however have an effect for
 10 simulations generated under *M0*, with higher average values obtained for simulations starting from North
 11 America (247 ± 85 vs. 79 ± 20 species).

12 In addition, successful simulations starting from North America generally started from an ancestor
 13 located closer to the isthmus of Panama than those starting from South America (Figure 3 C), again
 14 except for simulation generated under *M3*. Also, we can highlight a general trend that ancestral species
 15 originated closer to the isthmus in the case of successful simulations compared to unsuccessful ones
 16 (Figure S7-8). Interestingly enough, the results obtained for this metric, just like those obtained for the
 17 other two, hold for simulations generated after standardising the initial distances to the isthmus of North
 18 American virtual species with those starting from South America (Figure S4).

19 Biome selectivity of the simulated faunal exchanges

20 Due to the unequal area occupied by each biome between the two continents, the number of
 21 simulations starting within each biome was largely uneven across biome classes and continents (Figure 4
 22 B). The majority of North American ancestral species originated in the cold biome (57%), whereas the
 23 tropical biome recorded the largest proportion of ancestral species originating in South America (58%).
 24 Standardising initial distances to the isthmus of Panama for simulations originating in North America
 25 increased the proportion of simulations originating under low latitudes, to the detriment of those
 26 originating in high latitude, less represented in South America (Figure 4 A and Figure S5). Consequently,
 27 this increased the representativity of the tropical (from 5% to 10%) and arid (from 16% to 26%) biomes
 28 in the starting ranges of simulations originating in North America, and decreased the proportion of
 29 simulations originating in the cold (from 57% to 40%) and polar (from 4% to 2%) biomes (Figure S10).

30 Despite this initial imbalance, when simulations were generated under the *M0* model, our results
 31 suggest similar exchange intensity patterns between simulations starting from North and South America
 32 (Figure 4 C, first row). Fixed-niche simulations with tropical affinity (*i.e.*, originating in the tropical

1 biome and evolving under *M0*) were largely more frequently exchanged than fixed-niche simulations tied
 2 to another biome, with on average 90[77,100)% successful exchanges southwards (*i.e.*, from simulations
 3 originating in North America) and $66 \pm 5\%$ northwards (*i.e.*, from simulations originating in South
 4 America). Such proportions were significantly reduced when originating in the arid ($26 \pm 6\%$
 5 southwards, $18 \pm 9\%$ northwards) and temperate biomes ($26 \pm 10\%$ southwards, $29 \pm 8\%$ northwards).
 6 Almost no fixed-niche simulation originating in the cold or polar biome, no matter where it started,
 7 resulted in a successful inter-continental exchange. The exact same pattern is obtained for southwards
 8 dispersal events occurring after standardising initial distances to the isthmus of Panama (Figure S9, *M0*).

9 Regarding simulations generated under the adaptive-niche models (*M1–3*), those starting from
 10 North America maintained the same pattern of exchange intensities as simulations generated under *M0*.
 11 The proportion of exchange success was higher among those originating in the tropical biome (89
 12 [75,100)% for *M1*), followed by those originating in the temperate ($40 \pm 10\%$, *M1*) and arid ($34 \pm 11\%$
 13, *M1*) biomes. Also, almost no adaptive-niche simulation originating in the cold or polar biomes made it
 14 to South America (Figure 4 C *M1–2*). Note that removing the precipitation constraint on species climatic
 15 niche (*i.e.*, model *M3*) attenuated this imbalance (Figure 4 C *M3*). These results hold after standardising
 16 initial distances to the isthmus of Panama of simulations starting from North America to those starting
 17 from South America (Figure S9, *M1–3*). Conversely, the adaptive-niche disparity in proportion of
 18 successful exchanges across biomes was largely attenuated when simulations started in South America. In
 19 that case, successful exchanges appear almost equally-likely, regardless of the biome where simulations
 20 originate (Figure 4 C *M1–3*).

21 Discussion

22 An inverse asymmetry of exchange driven by climate

23 Throughout this work, we investigated the role of palaeoclimate in the asymmetry of the GABI
 24 through the use of eco-evolutionary simulations. Probably one of our most striking findings is that, under
 25 climatic constraints alone, simulated South American lineages dispersed to the North with a significantly
 26 greater intensity than the other way around (Figure 2). This therefore clearly contrasts with the long-
 27 recognised imbalanced success of North American mammal taxa compared to South Americans in the
 28 aftermath of the GABI (Simpson 1980). Seminal works primarily interpreted the empirically-derived
 29 asymmetry of the GABI under the lens of a continent-wide extension of MacArthur & Wilson's
 30 equilibrium theory (MacArthur & Wilson 1967). The latter theory establishes that, on evolutionary
 31 timescales, diversity within a given area should tend towards an equilibrium value proportional to the size
 32 of this area. Therefore, its continent-wide extension predicts the equilibrium diversity of North America

1 to be larger than that of South America, given the differences in size between the two continents (24 vs. $218 \times 10^6 \text{ km}^2$). Following this framework, Marshall *et al.* (1982) argued that the continent offering the 3 largest pool of taxa – which, according to this prediction, would be North America – would in turn have 4 resulted in the larger number of species exchanged in the end. It might be noticed that species pool 5 imbalance between two source areas has been identified as a key driver in the faunal exchange asymmetry 6 that contributed to the build-up of the unique biodiversity along the Wallace line (Skeels *et al.* 2023b). 7 Nonetheless, following such a reasoning, the probability for a given taxon to cross the Panamanian 8 landbridge is assumed to be independent from the side of the isthmus where it came from, which, 9 interestingly, is not supported by our findings (Figure 2). In addition, this theory does not align with the 10 actual imbalance in extant mammal species-level diversity between the Americas, being way larger in 11 South than in North America (Raven *et al.* 2020).

12 Although the inverse asymmetry in climate-driven exchange across both sides of the Isthmus 13 suggested by our findings goes in contradiction with the mammal fossil record (*e.g.*, Marshall *et al.* 14 1982), it actually echoes inter-American exchange patterns suggested by other groups of vertebrates. For 15 instance, the extant South American amphibian fauna is almost entirely endemic (Lomolino *et al.* 2006). 16 By contrast, the North American frog fauna is composed of four families of southern origin, and only 17 three of northern origin (Lomolino *et al.* 2006). Just like herpetofauna, tropical North American 18 ichthyofauna has long been recognised as mainly composed of groups that originated in South America 19 and further dispersed to the North, although most likely multiple dispersal events occurred, some of 20 which pre-dating the closure of the isthmus (*e.g.*, Chakrabarty & Albert 2011; Lomolino *et al.* 2006; 21 Smith *et al.* 2012). It should however be noted that, due to the requirement for connected freshwater 22 networks, the magnitude of the faunal admixture among freshwater fishes has been greatly limited 23 compared to land groups. Next, despite the higher dispersal abilities of birds, phylogenetic evidence has 24 shown a significant influence of the closure of the Panamanian Seaway in the rates of exchange between 25 both continent's avifaunas (Smith & Klicka 2010; Weir *et al.* 2009). In particular, these studies suggested 26 that transcontinental exchanges occurred symmetrically among clades inhabiting tropical regions, but 27 interestingly, just like land mammals, diversification of extra-tropical groups to tropical regions and 28 further dispersal to the other landmass was strongly asymmetrical in favour of North American groups 29 (Smith *et al.* 2012; Smith & Klicka 2010). That being said, notable exceptions of South American bird 30 groups that successfully diversified in North America (including within temperate regions) exist, such as 31 crown hummingbirds (*e.g.*, Wallace 1878). However, the lack of fossil record for all the aforementioned 32 groups may hamper the understanding of the processes behind their biogeographic history. Last, the 33 Central American flora has been characterised as a mixture of lineages of North and South American 34 origins, but phylogenetic evidence has suggested that the arrival of taxa from South America largely 35 predated the established timeline of the GABI (*e.g.*, Cody *et al.* 2010). Nevertheless, the orchid flora of

1 the Chocó region, bridging the north-western coast of South America and southern Panama, was found to
 2 be primarily made up with Andean – thus, South American – lineages that radiated locally not earlier
 3 than 5 million years ago (Pérez-Escobar *et al.* 2019).

4 **Revisiting climatic paradigms of the Great American Biotic Interchange**

5 One of the predictions of Vrba's resource-use hypothesis, formulated within the framework of her
 6 Habitat Theory and applied to the GABI, is that species with tropical affinity would have been naturally
 7 attracted towards the equator following the contraction of the tropical belt in the aftermath of the global
 8 cooling occurring at the Plio-Pleistocene boundary (Figure 1 B; Figure S1, from *ca.* 3 Ma). This would
 9 have actively promoted the establishment of North American tropical-suited taxa in South America (Vrba
 10 1992). Here, our virtual experiments partially corroborate these historical predictions. Indeed, fixed-niche
 11 simulations with tropical affinity of North American origin – coming from simulations starting in tropical
 12 North America and of which climatic niche could not evolve, model *M0* – were found to have a high
 13 prevalence of crossing the isthmus (Figure 4 C, model *M0*). However, the frequency of successful
 14 exchange obtained for fixed-niche tropical-suited simulations starting in South America was found to be
 15 almost equally-high, largely exceeding that of fixed-niche simulations tied to another biome and
 16 originating in the same continent (Figure 4 C, model *M0*). This goes against Vrba's seminal hypothesis,
 17 instead predicting that fewer mammals – only biome generalists or those suited to open habitats – from
 18 South America would be expected to roam the other way around in case of tropical contraction towards
 19 the equator. This finding could reflect an attenuated magnitude of the climatic fragmentation of the
 20 Panamanian region when compared to historical biome reconstructions (*e.g.*, Webb 1991), with the
 21 persistence of a tropical corridor in turn promoting the exchange of tropical specialists between both
 22 landmasses.

23 On average, one simulated radiation of open biome – *i.e.*, arid or temperate – specialists out of four
 24 resulted in a successful exchange, irrespective of its ancestral continent (Figure 4 C, model *M0*). This
 25 proportion is significantly lower than that obtained for tropical specialists. Yet, it has long been
 26 acknowledged that the migrant fossil record across both sides of the isthmus was essentially composed of
 27 taxa showing affinity with open biomes (*e.g.*, Vrba 1992). Among the most emblematic, we find for
 28 instance equids or camelids on the South American side, and ground sloths, toxodontids or glyptodontids
 29 on the North American side. The co-occurrence of distinct taxa in temporally well-delimited fossil
 30 assemblages led palaeontologists to postulate that the GABI was in fact a succession of exchange pulses
 31 punctuated by phases of interruption, further related to the climate-driven oscillations between savannah
 32 and rainforest in Central America during the Pleistocene (Webb 1991). Under such an assumption, the
 33 majority of exchanged taxa would therefore have ecologies suited to open habitats, and would have
 34 preferentially dispersed during Pleistocene glaciations (Bacon *et al.* 2016; Woodburne 2010) (Figure 1

1 A). As mentioned by Weir *et al.* (2009), the American mammal fossil record is highly biased, with few
 2 occurrences coming from low-latitude tropical regions, which is particularly striking in South America
 3 (e.g., Ugarte *et al.* 2025). Furthermore, evidence from extant mammal taxa suggest a strong tropical niche
 4 conservatism, that is, the trend according to which newly-originated tropical taxa would remain inhabiting
 5 under tropical climates (Smith *et al.* 2012; Wiens & Donoghue 2004). Thus, the fossil-derived number of
 6 mammal exchanges must have been greatly underestimated, with little fossil evidence of interchange
 7 constrained to the tropical biome, as stressed by Carrillo *et al.* (2020). Thus, one can wonder: could the
 8 empirically-derived migration imbalance in favour of North American taxa dispersing southwards be
 9 altered by the tropical niche conservatism of South-to-North migrants, thereby obscuring the actual
 10 number of intercontinental exchanges?

11 Our findings revealed that radiations following a colonisation of South America were more
 12 spatially spread than those arising from a northwards dispersal, except for simulations generated under the
 13 M3 model (Figure 3 A). The spatial spread of a migrant radiation was measured by the proportion of cells
 14 occupied within the newly-colonised continent relative to the size of the latter. When focusing on the
 15 absolute number of cells colonised – therefore without controlling for the area, in terms of number of
 16 cells, of the newly-colonised continent – such a finding only holds for successful simulations generated
 17 under M0 (Figure S11). This mirrors the fact that simulated tropical specialists from South America were
 18 more intensively exchanged than other simulated biome specialists (Figure 4 C, S6). The imposed tropical
 19 niche conservatism therefore constrained them to remain in tropical regions of North America, that are
 20 less represented in tropical North America than in South America (Figure 4 A). This adds some evidence
 21 supporting the fact that climate would have permitted South-to-North tropical exchanges to take place,
 22 but might have been obscured by a subsequent taphonomic bias in the fossil record, as suggested by
 23 studies relying on dated phylogenies (Bacon *et al.* 2015; Carrillo *et al.* 2020). Hence, our study calls for
 24 thinking about the distinction between the ability to disperse towards, and ability to maintain within the
 25 newly-colonised area. Those two distinct processes may have been ruled by different factors.

26 For additional remarks, limitations and perspectives, see *Supplementary text*.

27 In summary, our findings do not support the hypotheses exclusively relating the asymmetry of the
 28 GABI to palaeoclimate dynamics. Rather, our virtual experiments suggest that palaeoclimate would not
 29 have prevented reciprocal exchanges between the faunas of the two continents, especially within tropical
 30 areas. This stresses that the tropics did not act as a barrier as previously stated. Given the scarcity of fossil
 31 material in the tropical areas of the Americas, hypotheses relating the asymmetry of the GABI to
 32 palaeoclimate dynamics could have been affected by taphonomic bias, as they largely rely on the land
 33 mammal fossil record. This ultimately suggests that, although endemic land mammals from South
 34 America were virtually able to reach the other landmass, other unconsidered factors must have prevented

1 their establishment on the other continent, in turn shaping the widely-recognised asymmetry of this
2 textbook example of biotic interchange.

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1 Figures

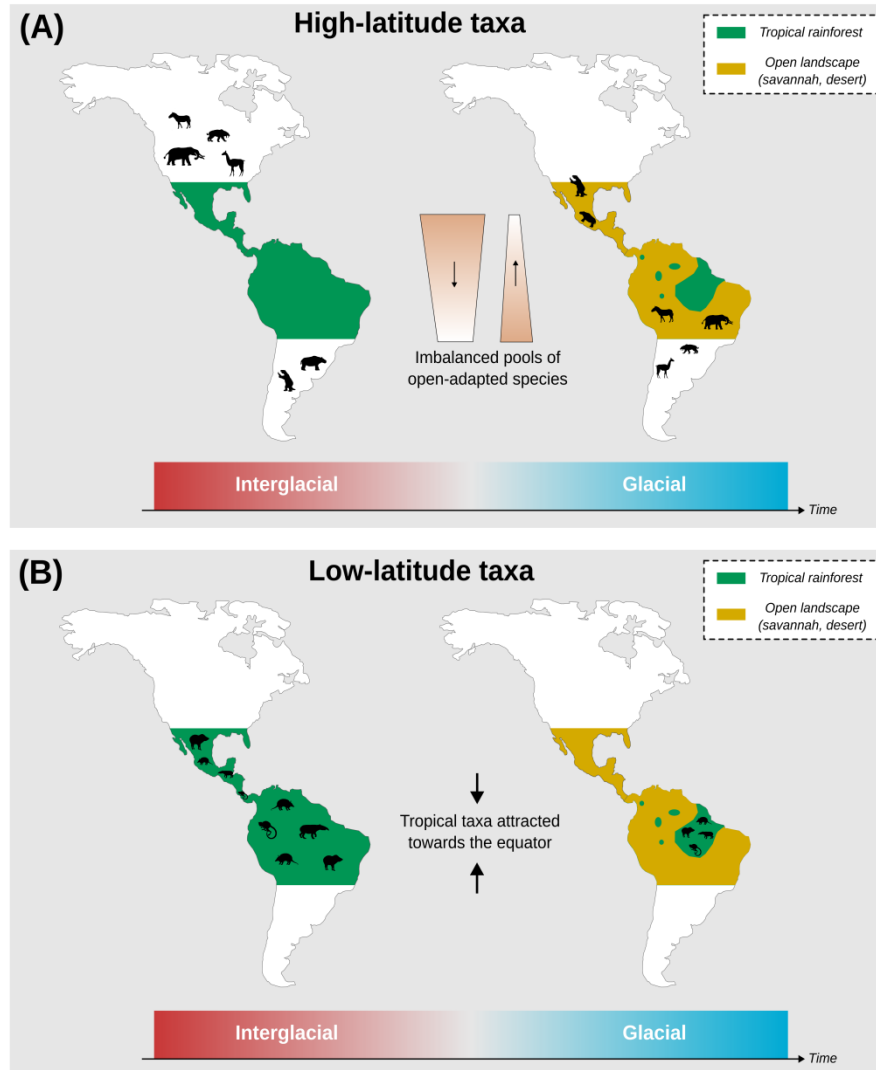


Figure 1: The two nonmutually-exclusive hypotheses relating the asymmetry of the GABI to the Pleistocene climatic oscillations, formulated in the framework of land mammals. Both involve biome transition from close tropical rainforest (green) during interglacial periods – warm and humid – to open savannah/desert (dark yellow) during glacial maxima – cold and dry – in the isthmian region. The first one (A), formulated by Webb (1991) views the asymmetry of the GABI as the result from the fact that North America had a higher pool of open-adapted taxa than South America. The opening of the isthmian region acted as an ecological filter favouring the dispersal of these open-adapted taxa to the detriment of the others. Hence, given the imbalance in open-adapted pool size between the two continents, more species would have been expected to disperse to the South than the other way around. The second one (B), formulated by Vrba (1992), proposes that the contraction of the tropical biome towards the equator triggered by the interglacial-glacial transition would have mechanically attracted tropical specialists southwards, irrespective of their continent of origin. Silhouettes are from phylopic, credits are available in Table S3. Note that the purpose of this figure is to illustrate our working hypotheses. Thus, we do not pretend these maps to accurately reflect inter-American biome dynamics between glacial and interglacial

periods. A more accurate picture of these dynamics is available on Figure S13.

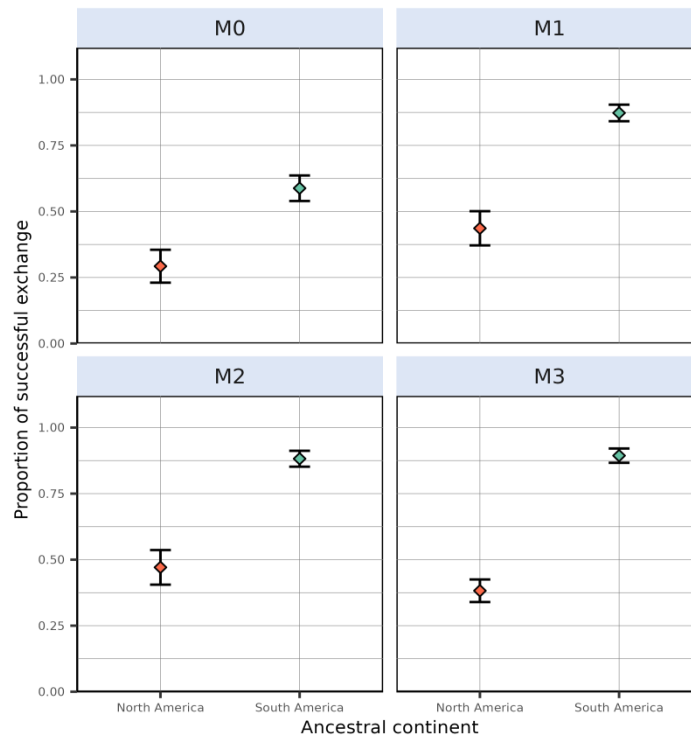


Figure 2: Proportion of simulations leading to a successful exchange to the other continent, computed as the ratio, given an initial continent and a modelling scenario, between the number of simulations leading to a transcontinental exchange and the total number of simulations resulting in at least one living species at the final state. From left to right are depicted the outcome of the fixed-niche model (*M0*, species climatic niche does not change through time), adaptive-niche model (*M1*, species climatic niche evolves following local changes in temperature and precipitations) and alternative versions of *M1* where species ecological niche is only determined by precipitation (*M2*) or temperature (*M3*). Each individual plot shows the outcome of simulations starting in North (red) and South (cyan) America. Dots represent mean proportion estimates and error bars show their associated Binomial-derived 95% confidence interval.

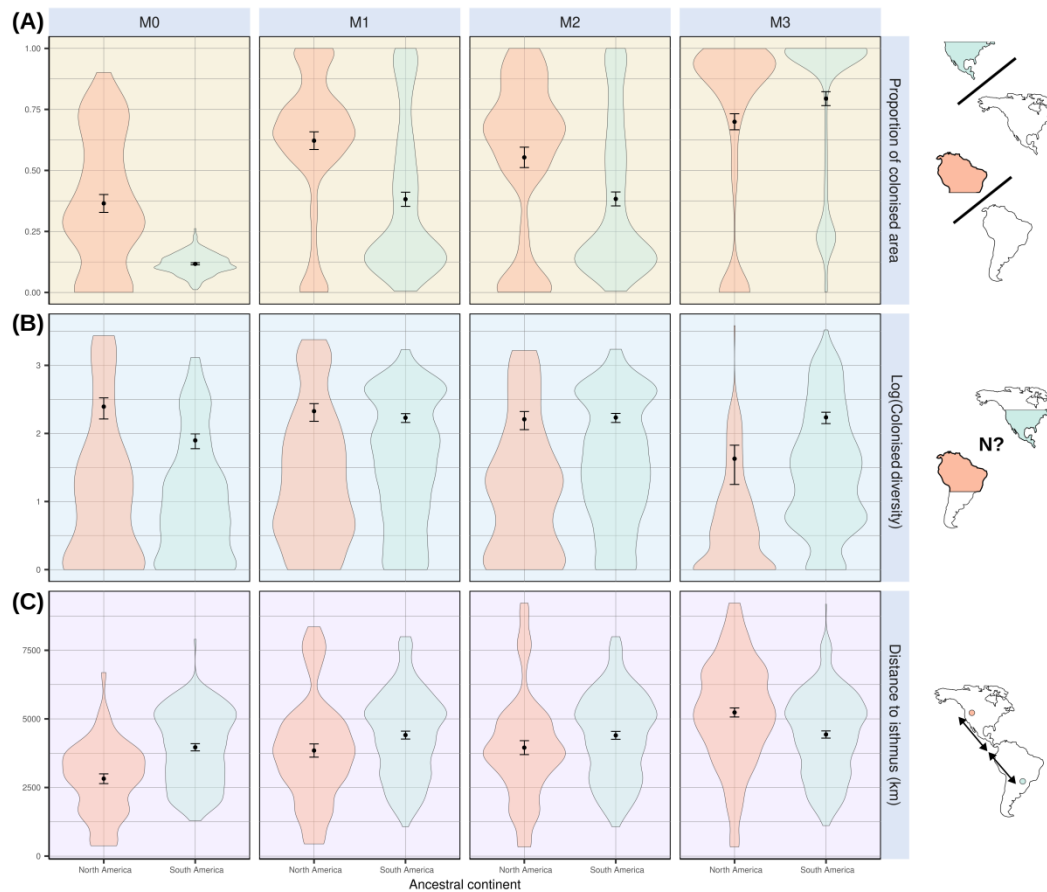


Figure 3: A few metrics summarising climate-mediated interchange given our four modelling schemes. These metrics were assessed for successful simulations only, i.e., simulations that resulted in a transcontinental faunal exchange. **(A)** Proportion of continent's newly-colonised area, quantified as the ratio between the number of pixels recording a presence in the landmass to colonise and the total number of pixels offered by this landmass. **(B)** Colonised diversity, expressed as the natural logarithm of the number of lineages within the colonised area. **(C)** Distance between simulation starting pixel and the isthmus of Panama. Mean estimates and their associated Student-derived 95% confidence intervals are displayed over their distribution.

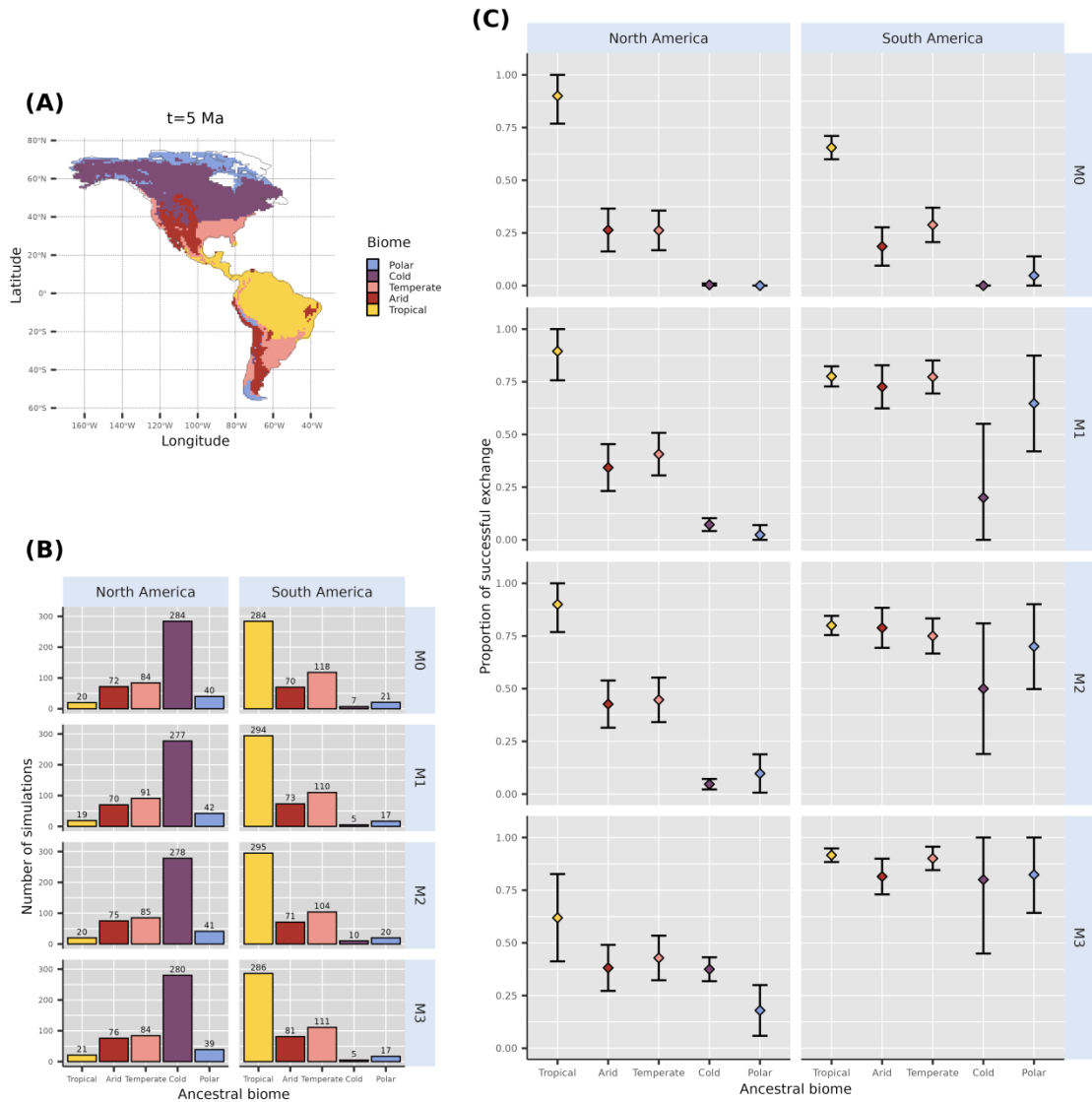


Figure 4: Initial biome where each ancestral species leading to successful simulations is recorded at the time of the onset of the simulations (5 Ma). **(A)** Global main climate zones based on Köppen-Geiger biome classification at $t = 5$ Ma. **(B)** Biome where the ancestral species of each of the 500 simulations originated, for each continent and each scenario. We removed simulations starting from a cell that could not be assigned a biome from our counts, hence the reason why some are not exactly summing to 500. **(C)** Proportion of successful simulations based on the biome where ancestral species originates, for each modelled scenario-ancestral continent combination. This proportion is expressed as the ratio between the number of successful simulations and the total number of simulations starting from a given biome. Coloured dots and error bars respectively show proportion estimates and their associated Binomial-derived 95% confidence intervals.

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Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

Lucas Buffan^{1*}, Fabien L. Condamine¹, [Philip B. Holden²](#), Sara Varela^{3§}

¹ Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France;

² [Environment, Earth and Ecosystem Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK](#)

^{3§} Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

* Corresponding author: lucas.l.buffan@gmail.com

Statement of authorship

S.V. and L.B. designed the project. [P.B.H. gave inputs on palaeoclimate data usage](#). L.B. ran the simulations, performed the analyses, produced the figures and wrote the manuscript with inputs from all co-authors.

Data availability statement

R scripts for generating, analysing and visualising the data are available at <https://github.com/Buffer3369/GABI-gen3sis.git>. Simulation landscapes and config files are available at <https://doi.org/10.6084/m9.figshare.31160704>.

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1 Abstract

2 Asymmetric colonisation patterns often arise from large-scale biotic exchanges involving two
 3 source areas. ~~ca. 5 million years ago~~In the Pliocene, the closure of the ~~isthmus of Panama~~ ~~nian-seaway~~
 4 facilitated the admixture between North and South American faunas, an event termed Great American
 5 Biotic Interchange (GABI). The ecological mechanisms underlying the asymmetry of the GABI,
 6 incontestably in favour of North American lineages, remain debated. Here, coupling a mechanistic eco-
 7 evolutionary simulator with palaeoclimatic reconstructions, we test fundamental hypotheses relating this
 8 biogeographic imbalance to palaeoclimate dynamics. When climate is the only constraint to species'
 9 dispersal, our simulations yield more South-to-North than North-to-South dispersal events, mostly within
 10 the tropical biome. Also, our models fail to explain the migrant diversity disequilibrium between the two
 11 continents, as well as temperate specialists' over-representation in the fossil record of successfully-
 12 exchanged taxa. Hence, we argue that spatio-temporal palaeoclimate dynamics ~~alone~~ cannot explain the
 13 asymmetry of the GABI, suggesting additional factors fashioned this macroecological enigma.

1 Introduction

2 The Great American Biotic Interchange (GABI) refers to the large-scale biotic exchange that
 3 occurred between the Americas and intensified after the closure of the isthmus of Panama (Marshall *et al.*
 4 1982; Stehli & Webb 1985; Webb 1976). Though the exact timing of the appearance of such a landbridge
 5 is not entirely resolved (Bourke *et al.* 2025; Jaramillo 2018; Montes *et al.* 2015; O’Dea *et al.* 2016),
 6 palaeontological evidence and molecular phylogenies both agree with a late Miocene onset of (non-
 7 volant) biotic exchange between the Americas (Carrillo *et al.* 2015; Tedford *et al.* 2004; Verzi *et al.*
 8 2008), with effective exchange pulses occurring during the Plio-Pleistocene (Bacon *et al.* 2015, 2016;
 9 Marshall *et al.* 1982; Woodburne 2010). Among land mammals, a puzzling feature of the GABI is that it
 10 impacted ~~both the North and South~~ continental faunas very differently. ~~In fact,~~ Although evidence from the
 11 fossil record indicates that the early phases of the GABI were marked by reciprocal land mammal
 12 exchanges between the Americas (Marshall *et al.* 1982; Webb 1976; Woodburne 2010), ~~However,~~ during
 13 the Pleistocene, the ability of migrant lineages to maintain and diversify in the newly-colonised continent
 14 became greater in South America (Woodburne 2010), ~~and m~~ Many native South American lineages
 15 concomitantly declined, some of which even going totally extinct (Boscaini *et al.* 2025; Carrillo *et al.*
 16 2020; Cione *et al.* 2015; Croft *et al.* 2020; Webb 1991). Hence, this intercontinental faunal exchange was
 17 asymmetric.

18 Although this macroecological imbalance is unquestionable, no clear consensus regarding the
 19 drivers underlying this asymmetry ~~yet~~ has been established ~~yet~~. On the one hand, biotic interactions
 20 involving migrant and native species have often been proposed to explain such an asymmetry. These
 21 would include competition (Domingo *et al.* 2020; Simpson 1980; Webb 1976) or predation of predator-
 22 naive preys (Faurby & Svenning 2016; Fernández-Monescillo *et al.* 2025; Marshall & Cifelli 1990; Webb
 23 1976). However, their contribution to the asymmetry of the GABI has been increasingly called into
 24 question (*e.g.*, Freitas-Oliveira *et al.* 2024a; Marsà *et al.* 2025; Prevosti *et al.* 2013; Tarquini *et al.* 2022;
 25 Vrba 1992). On the other hand, several studies ~~also~~ proposed ~~the role of abiotic factors—i.e., related to~~
 26 ~~species’ physical environment, such as climate~~ that the asymmetry of the GABI emerged from
 27 ~~palaeoclimate dynamics~~ (*e.g.*, Bacon *et al.* 2016; Freitas-Oliveira *et al.* 2024b; 2025; Marshall & Cifelli
 28 1990; Vrba 1992; Webb 1991). The prevailing views consist of relating pulses of transcontinental
 29 mammal exchanges with Pleistocene climatic oscillations, promoting the transition from tropical
 30 rainforest to savannah-like environments in Central America as climate cooled down (Figure 1).
 31 ~~Motivated by~~ Based on consideration of the migrant fossil record across both sides of the isthmus of
 32 Panama, Webb (1991) proposed that this climate-driven landscape opening would have facilitated the
 33 dispersal of open-adapted taxa. As those were assumed to be more numerous in North than in South
 34 America – the former continent offering more high-latitude temperate-to-cold environments than the latter

1—, a larger number of North American taxa would be predicted to disperse to the South than the other way
 2 around (Figure 1 A). The resulting exchanges would have been periodically interrupted by intervals of
 3 warmings-climate-warmed-up, promoting the reverse transition to rainforest in the isthmian region, with
 4 rainforests acting as a barrier for species dispersal (tropical ‘holding pen’ hypothesis) (Woodburne 2010).
 5 In addition, Vrba (1992) proposed the contraction of tropical habitats towards the equator promoted by
 6 the aforementioned periodical climatic cooling to have mechanically attracted taxa suited to tropical
 7 environments southwards (Figure 1 B). However, studies supporting these views have only been
 8 descriptive, tackling the issue from a pattern-oriented perspective driven by fossil data but, to our
 9 knowledge, never from a process-oriented perspective.

10 Recent improvements in process-explicit modelling have marked a turning point in the study of
 11 biodiversity (Pilowsky *et al.* 2022). By combining a landscape including palaeoenvironmental variables
 12 and a set of eco-evolutionary rules according to which virtual species are simulated, mechanistic eco-
 13 evolutionary models offer the unique opportunity to causally explore how biodiversity patterns arise and
 14 are maintained through evolutionary times (Hagen 2023). Recently, this class of models has provided
 15 unprecedented insights into the underlying forces behind widely-recognised biodiversity patterns, such as
 16 the latitudinal biodiversity gradient (Hagen *et al.* 2021a; Lorcery *et al.* 2025), the diversity disequilibrium
 17 between tropical rainforests across the globe (Hagen *et al.* 2021b), the build-up of the unique biodiversity
 18 along the Wallace line (Skeels *et al.* 2023b) and many others (*e.g.*, Boschman *et al.* 2023; Skeels *et al.*
 19 2023a).

20 Here, we test Webb and Vrba’s hypotheses on the role of palaeoclimate dynamics in the asymmetry
 21 of the GABI (Figure 1) using a spatially-explicit mechanistic biodiversity simulator coupled with
 22 palaeoclimate reconstructions across the last 5 million years. Our simulations tracked the diversification
 23 of a clade originating either in North or South America. The ability of a virtual species to maintainsurvive
 24 at a given place was only mediated by climatic factors. We hypothesise that (i) simulations starting from a
 25 North American ancestor would be more successful at reaching the other continent than the other way
 26 around, (ii) successful simulations starting from a North American ancestor would result in a more
 27 diverse and spatially spread radiation in the newly-colonised continent, ~~and~~ (iii) successful-interechange
 28 would be more frequently achieved among biome-generalists, extratropical-specialists rather than tropical
 29 specialists.niche evolvability and (iv) initial biome would significantly impact exchange frequencies, with
 30 higher values obtained for adaptive-niche simulations, or fixed-niche simulations with temperate or arid
 31 rather than tropical affinity.

32

1 Material and Methods

2 Beforehand, it has to be noticed that, across the entire study, ‘North America’ actually refers to the
3 current North and Central American continents. The delimitation between North and South America was
4 arbitrarily set as a straight line connecting two points located along the eastern and western coasts of
5 present Panama (82.882° W, 7.138° N and 81.369° W and 9.008° N) (see Figure S1).

6 Palaeoclimate data

7 We used the spatio-temporal terrestrial palaeoclimate series ~~from~~of the PALEO-PGEM emulator
8 (Holden *et al.* 2019) of the PLASIM-GENIE intermediate-complexity atmosphere-ocean general
9 circulation model (Holden *et al.* 2016). This reconstruction coversing the last 5 Million years (My)
10 with downscaled to a $1 \times 1^\circ$ spatial resolution (Barreto *et al.* 2023). For the scope of this work, we used
11 two bioclimatic variables: mean annual temperature (MAT, BIO1) and mean annual precipitation (MAP,
12 BIO12). To avoid ~~reaching an overkillingly intractably~~-high number of simulation time steps, we reduced
13 the temporal resolution of the climate simulations from a 1,000-years to a 10,000-years time step by
14 sampling corresponding time slices. This procedure did not alter the overall climatic trends. In particular,
15 Pleistocene climatic oscillations clearly remained after downscaling (Figure S12).

16 ~~Additionally, in~~ order to discriminate simulations based on the biome where the initial species
17 originates, we reclassified monthly palaeoclimate layers for ~~our~~the starting time ($t = 5$ My ago)
18 according to Köppen-Geiger broad climate types (Beck *et al.* 2018; Galván *et al.* 2023). This
19 classification divides the world into five broad climatic regions: Polar, Cold, Temperate, Arid and
20 Tropical, following the criteria proposed by Köppen (~~Köppen~~-1900).

21 Simulation model and modelling schemes

22 We performed spatially-explicit simulations of biodiversity using gen3sis v1.5.11 (Hagen *et al.*
23 2021a). Simulations, moving forward-in-time with a 10ky time step, tracked the radiation of a clade
24 arising from a single ancestral population, either originating in North or in South America, throughout the
25 last 5 My. In gen3sis, virtual species are modelled with individual populations evolving independently,
26 following a set of four rules: dispersal, speciation, environmental filtering and trait evolution. All
27 simulation scenarios that we implemented only differed on the last rule (trait evolution), the first three
28 being common to all models (~~or at least partly~~).

29 We elaborated a gridded landscape ~~relying on~~from the ~~two~~ time-series of the ~~aforementioned two~~
30 palaeoclimatic layers. Within this landscape, populations making up virtual species inhabited a set of
31 connected cells. At each time step, inter-cell dispersal could occur with an intensity sampled in a Weibull
32 distribution (repeated independently for each species-cell combination). As done by Skeels *et al.* (2023a;

2023b), the cost of moving from a cell to another was made twice harder if the target cell was not habitable (*i.e.*, ocean or sea). If, by chance, two populations happened to become disconnected, they would accumulate divergence as long as they get remain disconnected, at a rate ruled by a divergence factor, here set to 0.05 (Table S1). Once passed a divergence threshold, speciation would occur – note that, consequently, the only mode of speciation allowed by gen3sis is allopatry.

As our goal was to evaluate the extent to which palaeoclimate change may have shaped the unequal success of North American taxa establishing southwards compared to South American taxa establishing northwards, local climate was set as the only constraint for a population to maintain survive in a given cell. Each species was therefore assigned a two-dimensional climatic niche, one axis corresponding to temperature suitability and the other to precipitation suitability. The abundance of a species along an environmental gradient was ruled determined by two parameters – that we also call traits – : a population- and species-specific optimum (resp. $P_{k,l}$ and $T_{k,l}$ for the precipitation and temperature optima of the population k of the species l) and a breadth (or tolerance) (resp. ω_P and ω_T ; within a simulation, breadths were shared across all populations and species had the same breadth). A population could maintain survive in a given cell only if the difference between its optima and local climate (both temperature and precipitation) was below less than its breadth tolerance. Therefore, given a site i with local precipitation and temperature at time t being $P_i(t)$ and $T_i(t)$, this translates in

$$\begin{cases} |P_i(t) - P_{k,l}| < \omega_P \\ |T_i(t) - T_{k,l}| < \omega_T \end{cases}$$

Each simulation started with a single ancestral species. The range of this ancestral species was randomly selected within the desired ancestral area (*i.e.*, North or South America). To avoid oversampling high latitudes, we merged the ancestral landscape with the centroids of an equal-area grid (~100km spacing) generated using the R package *h3jsr* (O'Brien 2023). We then sampled a centroid at random and retrieved the corresponding cell in our landscape (Figure S5 left and right). To reduce the probability of initial crash, the climate optima parameters of the ancestor were set to those of the local environment. However, due to the geometry of the two continents, this sampling scheme resulted in ancestral species starting from South America originating on average closer to the isthmus of Panama than those from North America (Figure S2 top). To control for this spatial bias, we assessed the distance to the isthmus of each cell of the North American portion of our $1^\circ \times 1^\circ$ landscape. We then carried out other batches of simulations starting from North America, constraining the initial distance to the isthmus of the ancestral species of the simulation i to be as close as possible to the same distance involving the ancestral species of the simulation i starting from South America (Figure S2 bottom, Figure S5 middle).

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1 We implemented various simulation scenarios (also called modelling schemes) where we
 2 modulated the ability of species to cope with climate change. In the adaptive niche model (*M1*), climatic
 3 optima ($P_{k,l}$ and $T_{k,l}$) were able to evolve in the direction of local climate change. To isolate the effect of
 4 a single environmental covariate on the resulting simulated biodiversity pattern, alternative versions of
 5 this model consisted of either removing the temperature (*M2*) or the precipitation (*M3*) constraint on
 6 species ecology. Besides, we also implemented a fixed niche model (*M0*), where niche optima could not
 7 evolve and were therefore held constant (to their initial value) across all simulations. 500 replicates were
 8 made under member simulation ensembles were performed for each simulation scenario each modelling
 9 scheme (detailed in the next paragraph), and for both with North and South America as ancestral area. For
 10 details about initialisation of a simulation and parameter values, please refer to *Supplementary text*.

11 Across replicates, we varied 6 key parameters: (1) the divergence threshold above which speciation
 12 occurred (S , range = [1,3]), (2) the rate of environmental niche evolution — only in models where we
 13 allowed climatic niche to evolve (σ_e , [0.005,0.15]), (3) the shape (ϕ , [0.5,3]) and (4) scale (ψ , [0.1,1])
 14 of the Weibull distribution acting as a dispersal kernel, (5) the strength of environmental filtering (*i.e.*,
 15 niche breadth) for temperature (ω_T ,) and (6) precipitation (ω_P , [0.05,0.3]). ω_T As done by many authors,
 16 we ensured an even sampling of the resulting multidimensional parameter space via a quasirandom
 17 sampling technique relying on Sobol' sequences (*e.g.*, Hagen *et al.* 2021b; Skeels *et al.* 2023b). Parameter
 18 ranges were determined on the basis of a literature review, ensuring their biological relevance. These
 19 ranges were further adjusted to limit the prevalence of crashing simulations. In addition to these varying
 20 parameters, we specified the value of several fixed parameters (Table S1).

21 We implemented various simulation scenarios (also called modelling schemes) where we
 22 modulated the ability of species to cope with climate change, and thus, the generalist vs. specialist nature
 23 of the simulated species. In the full adaptationist model (*M1*, generalist), climatic optima ($P_{k,l}$ and $T_{k,l}$)
 24 were able to evolve in the direction of local climate change. To isolate the effect of a single
 25 environmental covariate on the resulting simulated biodiversity pattern, alternative versions of this model
 26 consisted of either removing the temperature (*M2*) or the precipitation (*M3*) constraint on species
 27 ecology. Besides, we also implemented a zero adaptationist model (*M0*, specialist), where niche optima
 28 could not evolve and were therefore held constant (to their initial value) across all simulations.

29 Importantly, simulations generated under the fixed niche model (*M0*) originating in the tropical
 30 biome of either continent allowed to assess fundamental hypotheses related to the concept of tropical
 31 niche conservatism (Wiens & Donoghue 2004). This concept has often been invoked to explain the high
 32 diversity of tropical compared to temperate regions. In the framework of the Americas, Smith *et al.*
 33 (2012) evidenced that taxa of North American origin were a significant compound of the high Neotropical

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biodiversity, whereas taxa with a southern ascent were little represented in temperate North America, therefore arguing that South American taxa exhibit an higher niche conservatism (in favour of tropical areas) than North American taxa. Thus, our modelling schemes also provide the ability to test whether palaeoclimate is a sufficient driver behind this observed niche conservatism asymmetry across the Americas.

Throughout the study period, a vast proportion the South American landmass was dominated by relatively stable tropical climatic conditions whereas the overall North American climate was dominated by temporal fluctuations (Figure S1). Consequently, we expect that lineages generated under fixed-niche simulations (M0, high conservatism) originating in South America to be less intensively exchanged than those generated under evolving-niche simulations starting in North America (M1–3, attenuated conservatism). Such an imbalance would be in line with the empirical pattern of exchange asymmetry that took place during the GABI.

Last, it is important to notice that, since the palaeoclimate reconstructions that we used ignored tectonic plate motion – a reasonable assumption for a 5-My time frame – continental position in our landscapes was held constant across simulations. Therefore, in that framework, changes in connectivity between North and South America could only arise from the environmentclimate.

Quantification and statistical analyses

At every time step, for each run of each simulation, we used a function released by Skeels *et al.* (2023b) to save presence-absence matrices. Given a time t , such matrices stored on their columns the presence of each simulated species that have existed from the onset of the simulation to t within each cell of the landscape (rows of the matrix) as a binary variable – 1: the species inhabits that cell, 0: it does not.

Given a batch of simulations, after discarding those that crashed – *i.e.*, that led to no living representative – we quantified four metrics reflecting the occurrence, magnitude and efficiency of an eventual transcontinental exchange based on the presence-absence matrix of the last time step of each successful run (hereafter called t_{end}). First, we assessed the exchange success, a binary variable set to one if at least one pixel of the area to colonise (hereafter called A_{col} , *e.g.*, South America if the simulation started in North America) recorded a presence. Then, we recorded the biome in which the initial ancestor occurs (*i.e.*, Tropical, Arid, Temperate, Cold, Polar) as well as the distance of this initial ancestor to a point arbitrarily located on the zone of the isthmus of Panama, (-84°lon , 10°lat). Last, in the case of a successful exchange, we quantified the number of virtual taxa recorded in A_{col} at t_{end} – referred to as “colonised diversity” – and the proportion of A_{col} actually colonised at t_{end} as the ratio between the number of cells occupied by the colonisers and the total number of available cells (*i.e.*, surface of A_{col}).

1

1 Results

2 South-to-North dispersal events are made easier by climate

3 Our results suggest that the proportion of simulations successfully reaching the other continent is
 4 always higher – by approximately a factor of two – when starting from South America than from North
 5 America (Figure 2). This imbalance holds whether we allow species climatic niche to evolve following
 6 the direction of local climatic change ($44 \pm 6\%$ vs. $87 \pm 3\%$, model *M1*) or not ($29 \pm 6\%$ vs. $59 \pm 5\%$,
 7 *M0*), although we can notice that overall transcontinental exchange frequencies are lower in the latter
 8 case. When allowing climatic niches to evolve, amputation of either their temperature ($47 \pm 7\%$ vs. 88
 9 $\pm 3\%$, *M2*) or precipitation ($38 \pm 4\%$ vs. $89 \pm 3\%$, *M3*) compounds did not alter the pattern. We
 10 emphasise the fact that, contrary to our expectations, fixed-niche simulations originating in South
 11 America always yielded to a greater exchange intensity than evolving-niche simulations starting in North
 12 America. This imbalance was also The simulated imbalance maintained/persisted when/after standardising
 13 initial distances to the isthmus of Panama for simulated species originating in North America to those of
 14 simulated species from South America, except for simulations generated under the zero-adaptationist/fixed
 15 niche model (*M0*, $50 \pm 6\%$ vs. $59 \pm 5\%$) (Figure S4). Last, we stress that the simulated pattern was also
 16 robust to thermal breadth expansion for lineages originating in North America, up to 10 times broader
 17 than for simulations originating in South America (Figure S14 A).

18 Out of the 500 runs carried out for each initial landmass-modelling scenario combination, the
 19 proportion of simulations leading to no living representative (=crash) was on average three times higher
 20 for simulations starting from North America, except for model *M3*, recording almost no crash (Figure S14
 21 B, Table S2). This difference, although attenuated, persisted after the spatial standardisation procedure,
 22 but fully disappeared when we increased the thermal breadth of simulations originating in North America
 23 (Figure S14).

24 Successful establishment on the newly-colonised continent

25 Irrespective of the initial continent, our findings highlight that successful simulations generated
 26 under models allowing species climatic niches to evolve (*M1–M3*) colonised a larger area on the newly-
 27 colonised continent than those generated under the *M0* model – where climatic niche evolution was not
 28 allowed (Figure 3 A). Furthermore, species of North American origin that successfully made it to South
 29 America generally colonised a greater area than those of South American origin reaching North America
 30 (Figure 3 A). This pattern is particularly striking for simulations generated under the *M0* model ($36.5 \pm$
 31 3.5% vs. $11.7 \pm 0.4\%$), and holds for all but the *M3* model – relaxing the precipitation constraint on
 32 species' niche –, the latter suggesting an opposite trend ($69.9 \pm 3.3\%$ vs. $79.4 \pm 3.9\%$). However, due to
 33 the difference in size (*i.e.*, number of pixels) between the two continents (Figure S1), the actual colonised

area was greater when simulations generated under the *M1–3* models started from South than North America, but remained larger when simulations were generated under *M0* started from North America (Figure S11).

Besides, contrary to the proportion of colonised area for a given ancestral continent, colonised diversity does not vary significantly across our four models. However, a significant decrease is evident in simulations starting from North America generated under the *M3* model (Figure 3 B). Moreover, ancestral continent does not seem to affect mean colonised diversity values in the case of the *M1* (212 ± 86 vs. 170 ± 25 species respectively retrieved in the newly-colonised continent after starting from North or South America) and *M2* (161 ± 48 vs. 171 ± 27 species) models. It does however have an effect for simulations generated under *M0*, with higher average values obtained for simulations starting from North America (247 ± 85 vs. 79 ± 20 species).

In addition, successful simulations starting from North America generally started from an ancestor located closer to the isthmus of Panama than those starting from South America (Figure 3 C), again except for simulation generated under *M3*. Also, we can highlight a general trend that ancestral species originated closer to the isthmus in the case of successful simulations compared to unsuccessful ones (Figure S7-8). Interestingly enough, the results obtained for this metric, just like those obtained for the other two, hold for simulations generated after standardising the initial distances to the isthmus of North American virtual species with those starting from South America (Figure S4).

Biome selectivity of the simulated faunal exchanges

Due to the unequal area occupied by each biome between the two continents, the number of simulations starting within each biome was largely uneven across biome classes and continents (Figure 4 B). The majority of North American ancestral species originated in the cold biome (57%), whereas the tropical biome recorded the largest proportion of ancestral species originating in South America (58%). Standardising initial distances to the isthmus of Panama for simulations originating in North America increased the proportion of simulations originating under low latitudes, to the detriment of those originating in high latitude, less represented in South America (Figure 4 A and Figure S5). Consequently, this increased the representativity of the tropical (from 5% to 10%) and arid (from 16% to 26%) biomes in the starting ranges of simulations originating in North America, and decreased the proportion of simulations originating in the cold (from 57% to 40%) and polar (from 4% to 2%) biomes (Figure S10).

Despite this initial imbalance, when simulations were generated under the *M0* model, our results suggest similar exchange intensity patterns between simulations starting from North and South America (Figure 4 C, first row). ~~Tropical specialists~~ Fixed-niche simulations with tropical affinity (*i.e.*, originating

1 in the tropical biome and evolving under *M0*) were largely more frequently exchanged than ~~other-biome~~
2 ~~specialistsfixed-niche simulations tied to another biome~~, with on average 90[77,100)% successful
3 exchanges southwards (*i.e.*, from simulations originating in North America) and $66 \pm 5\%$ northwards
4 (*i.e.*, from simulations originating in South America). Such proportions were significantly reduced
5 ~~for when originating in the~~ arid ($26 \pm 6\%$ southwards, $18 \pm 9\%$ northwards) and temperate
6 ~~specialistsbiomes~~ ($26 \pm 10\%$ southwards, $29 \pm 8\%$ northwards), ~~and a~~ Almost no ~~fixed-niche simulation~~
7 ~~originating in the~~ cold or polar ~~specialistbiome~~, no matter where ~~theyit~~ ~~originatedstarted~~, ~~succcessfully~~
8 ~~made it to the other continentresulted in a successful inter-continental exchange~~. The exact same pattern is
9 obtained for southwards dispersal events occurring after standardising initial distances to the isthmus of
10 Panama (Figure S9, *M0*).

11 Regarding ~~biome-generalistssimulations generated under the adaptive-niche models~~ (~~models-M1-3~~
12), ~~simulationsthose~~ starting from North America maintained the same ~~distributionpattern~~ of exchange
13 intensities across biomes as ~~biome-specialistssimulations generated under M0~~. ~~Generalists that originated~~
14 ~~in the tropical biome record the highest p~~The proportion of exchange success ~~was higher among those~~
15 ~~originating in the tropical biome~~ ($89[75,100)\%$ for *M1*), followed by those originating in the temperate ($64 \pm 10\%$, *M1*) and arid ($34 \pm 11\%$, *M1*) biomes. ~~Also, generalistsAlso, almost no adaptive-niche~~
17 ~~simulation~~ originating in the cold or polar biomes ~~almost never~~ made it to South America (Figure 4 C *M1*
18 –2). Note that removing the precipitation constraint on species climatic niche (*i.e.*, model *M3*) attenuated
19 this imbalance (Figure 4 C *M3*). ~~All this still holdsThese results hold whenafter~~ standardising initial
20 distances to the isthmus of Panama of simulations starting from North America to those starting from
21 South America (Figure S9, *M1-3*). Conversely, the ~~adaptive-niche~~ disparity in proportion of successful
22 exchanges across biomes was largely attenuated ~~for biome-generalistswhen originatingsimulations started~~
23 in South America, ~~for which-In that case, a~~successful exchanges appears almost equally-likely,
24 regardless of the biome where ~~theysimulations~~ originate (Figure 4 C *M1-3*).

25 Discussion

26 An inverse asymmetry of exchange driven by climate

27 Throughout this work, we investigated the role of palaeoclimate in the asymmetry of the GABI
28 through the use of eco-evolutionary simulations. Probably one of our most striking findings is that, under
29 climatic constraints alone, simulated South American lineages dispersed to the North with a significantly
30 greater intensity than the other way around (Figure 2). This therefore clearly contrasts with the long-
31 recognised imbalanced success of North American mammal taxa compared to South Americans in the
32 aftermath of the GABI (Simpson 1980).

1 Seminal works primarily interpreted the empirically-derived asymmetry of the GABI under the lens
 2 of a continent-wide extension of MacArthur & Wilson's equilibrium theory (MacArthur & Wilson 1967).
 3 The latter theory establishes that, on evolutionary timescales, diversity within a given area should tend
 4 towards an equilibrium value proportional to the size of this area. Therefore, its continent-wide extension
 5 predicts the equilibrium diversity of North America to be larger than that of South America, given the
 6 differences in size between the two continents (24 vs. 18×10^6 km²). Following this framework,
 7 Marshall *et al.* (1982) argued that the continent offering the largest pool of taxa – which, according to this
 8 prediction, would be North America – would in turn have resulted in the larger number of species
 9 exchanged in the end. It might be noticed that species pool imbalance between two source areas has been
 10 identified as a key driver in the faunal exchange asymmetry that contributed to the build-up of the unique
 11 biodiversity along the Wallace line (Skeels *et al.* 2023b). Nonetheless, following such a reasoning, the
 12 probability for a given taxon to cross the Panamanian landbridge is assumed to be independent from the
 13 side of the isthmus where it came from, which, interestingly, is not supported by our findings (Figure 2).
 14 In addition, this theory does not align with the actual imbalance in extant mammal species-level diversity
 15 between the Americas, being way larger in South than in North America (Raven *et al.* 2020).

16 Although the inverse asymmetry in climate-driven exchange across both sides of the Isthmus
 17 suggested by our findings goes in contradiction with the mammal fossil record (*e.g.*, Marshall *et al.*
 18 1982), it actually echoes inter-American exchange patterns suggested by other groups of vertebrates. For
 19 instance, the extant South American amphibian fauna is almost entirely endemic (Lomolino *et al.* 2006).
 20 By contrast, the North American frog fauna is composed of four families of southern origin, and only
 21 three of northern origin (Lomolino *et al.* 2006). Just like herpetofauna, tropical North American
 22 ichthyofauna has long been recognised as mainly composed of groups that originated in South America
 23 and further dispersed to the North, although most likely multiple dispersal events occurred, some of
 24 which pre-dating the closure of the isthmus (*e.g.*, Chakrabarty & Albert 2011; Lomolino *et al.* 2006;
 25 Smith *et al.* 2012). It should however be noted that, due to the requirement for connected freshwater
 26 networks, the magnitude of the faunal admixture among freshwater fishes has been greatly limited
 27 compared to land groups. Next, despite the higher dispersal abilities of birds, phylogenetic evidence has
 28 shown a significant influence of the closure of the Panamanian Seaway in the rates of exchange between
 29 both continent's avifaunas (Smith & Klicka 2010; Weir *et al.* 2009). In particular, these studies suggested
 30 that transcontinental exchanges occurred symmetrically among clades inhabiting tropical regions, but
 31 interestingly, just like land mammals, diversification of extra-tropical groups to tropical regions and
 32 further dispersal to the other landmass was strongly asymmetrical in favour of North American groups
 33 (Smith *et al.* 2012; Smith & Klicka 2010). That being said, notable exceptions of South American bird
 34 groups that successfully diversified in North America (including within temperate regions) exist, such as
 35 [crown](#) hummingbirds (*e.g.*, Wallace 1878). However, the lack of fossil record for all the aforementioned

1 groups may hamper the understanding of the processes behind their biogeographic history. Last, the
 2 Central American flora has been characterised as a mixture of lineages of North and South American
 3 origins, but phylogenetic evidence has suggested that the arrival of taxa from South America largely
 4 predated the established timeline of the GABI (*e.g.*, Cody *et al.* 2010). Nevertheless, the orchid flora of
 5 the Chocó region, bridging the north-western coast of South America and southern Panama, was found to
 6 be primarily made up with Andean – thus, South American – lineages that radiated locally not earlier
 7 than 5 million years ago (Pérez-Escobar *et al.* 2019).

8 Revisiting climatic paradigms of the Great American Biotic Interchange

9 One of the predictions of Vrba's resource-use hypothesis, formulated within the framework of her
 10 Habitat Theory and applied to the GABI, is that species with tropical specialists affinity would have been
 11 naturally attracted towards the equator following the contraction of the tropical belt in the aftermath of the
 12 global cooling occurring at the Plio-Pleistocene boundary (Figure 1 B; Figure S1, from *ca.* 3 Ma). This
 13 would have actively promoted the establishment of North American tropical specialists suited taxa in
 14 South America (Vrba 1992). Here, our virtual experiments partially corroborate these historical
 15 predictions. Indeed, simulated tropical generalists fixed-niche simulations with tropical affinity of North
 16 American origin – coming from simulations starting in tropical North America and of which climatic
 17 niche could not evolve, model *M0* – were found to have a high prevalence of crossing the isthmus
 18 (Figure 4 C, model *M0*). However, the prevalence of tropical specialists frequency of successful exchange
 19 obtained for fixed-niche tropical-suited simulations starting in from South America to disperse to the
 20 North was found to be almost equally-high, largely exceeding that of other biome specialists fixed-niche
 21 simulations tied to another biome and originating in the same continent (Figure 4 C, model *M0*). This
 22 goes against Vrba's seminal hypothesis, instead predicting that fewer mammals – only biome generalists
 23 or open-habitat specialists those suited to open habitats – from South America would be expected to roam
 24 the other way around in case of tropical contraction towards the equator. This finding could reflect an
 25 attenuated magnitude of the climatic fragmentation of the Panamanian region when compared to
 26 historical biome reconstructions (*e.g.*, Webb 1991), with the persistence of a tropical corridor in turn
 27 promoting the exchange of tropical specialists between both landmasses.

28 On average, one simulated radiation of open biome – *i.e.*, arid or temperate – specialists out of four
 29 resulted in a successful exchange, irrespective of its ancestral continent (Figure 4 C, model *M0*). This
 30 proportion is significantly lower than that obtained for tropical specialists. Yet, it has long been
 31 acknowledged that the migrant fossil record across both sides of the isthmus was essentially composed of
 32 taxa showing affinity with open biomes (*e.g.*, Vrba 1992). Among the most emblematic, we find for
 33 instance equids, proboscids or camelids on the South American side, and ground sloths, toxodontids or
 34 glyptodontids on the North American side. The co-occurrence of distinct taxa in temporally well-

1 delimited fossil assemblages led palaeontologists to postulate that the GABI was in fact a succession of
 2 exchange pulses punctuated by phases of interruption, further related to the climate-driven oscillations
 3 between savannah and rainforest in Central America during the Pleistocene (Webb 1991). Under such an
 4 assumption, the majority of exchanged taxa would therefore have ecologies suited to open habitats, and
 5 would have preferentially dispersed during Pleistocene glaciations (Bacon *et al.* 2016; Woodburne 2010)
 6 (Figure 1 A). As mentioned by Weir *et al.* (2009), the American mammal fossil record is highly biased,
 7 with few occurrences coming from low-latitude tropical regions, which is particularly striking in South
 8 America (*e.g.*, Ugarte *et al.* 2025). Furthermore, evidence from extant mammal taxa suggest a strong
 9 tropical niche conservatism, that is, the trend according to which newly-originated tropical taxa would
 10 remain inhabiting under tropical climates (Smith *et al.* 2012; Wiens & Donoghue 2004). Thus, the fossil-
 11 derived number of mammal exchanges must have been greatly underestimated, with little fossil evidence
 12 of interchange constrained to the tropical biome, as stressed by Carrillo *et al.* (2020). Thus, one can
 13 wonder: could the empirically-derived migration imbalance in favour of North American taxa dispersing
 14 southwards be altered by the tropical niche conservatism of South-to-North migrants, thereby obscuring
 15 the actual number of intercontinental exchanges?

16 Our findings revealed that radiations following a colonisation of South America were more
 17 spatially spread than those arising from a northwards dispersal, except for simulations generated under the
 18 *M3* model (Figure 3 A). The spatial spread of a migrant radiation was measured by the proportion of cells
 19 occupied within the newly-colonised continent relative to the size of the latter. When focusing on the
 20 absolute number of cells colonised – therefore without controlling for the area, in terms of number of
 21 cells, of the newly-colonised continent – such a finding only holds for successful simulations generated
 22 under *M0* (Figure S11). This mirrors the fact that simulated tropical specialists from South America were
 23 more intensively exchanged than other simulated biome specialists (Figure 4 C, S6). The imposed tropical
 24 niche conservatism therefore constrained them to remain in tropical regions of North America, that are
 25 less represented in tropical North America than in South America (Figure 4 A). This adds some evidence
 26 supporting the fact that climate would have permitted South-to-North tropical exchanges to take place,
 27 but might have been obscured by a subsequent taphonomic bias in the fossil record, as suggested by
 28 studies relying on dated phylogenies (Bacon *et al.* 2015; Carrillo *et al.* 2020). Hence, our study calls for
 29 thinking about the distinction between the ability to disperse towards, and ability to maintain within the
 30 newly-colonised area. Those two distinct processes may have been ruled by different factors.

31 For additional remarks, limitations and perspectives, see *Supplementary text*.

32 In summary, our findings do not support the hypotheses exclusively relating the asymmetry of the
 33 GABI to palaeoclimate dynamics. Rather, our virtual experiments suggest that palaeoclimate would not
 34 have prevented reciprocal exchanges between the faunas of the two continents, especially within tropical

1 areas. This stresses that the tropics did not act as a barrier as previously stated. Given the scarcity of fossil
 2 material in the tropical areas of the Americas, hypotheses relating the asymmetry of the GABI to
 3 palaeoclimate dynamics could have been affected by taphonomic bias, as they largely rely on the land
 4 mammal fossil record. This ultimately suggests that, although endemic land mammals from South
 5 America were virtually able to reach the other landmass, other unconsidered factors must have prevented
 6 their establishment on the other continent, in turn shaping the widely-recognised asymmetry of this
 7 textbook example of biotic interchange.

8 Acknowledgements

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1 Figures

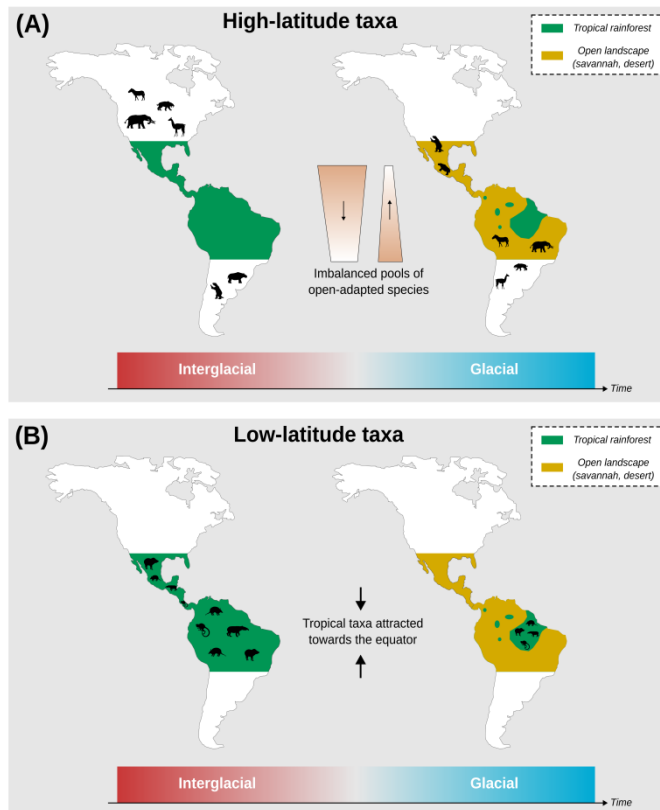


Figure 1: The two nonmutually-exclusive hypotheses relating the asymmetry of the GABI to the Pleistocene climatic oscillations, formulated in the framework of land mammals. Both involve biome transition from close tropical rainforest (green) during interglacial periods – warm and humid – to open savannah/desert (dark yellow) during glacial maxima – cold and dry – in the isthmian region. The first one (A), formulated by Webb (1991) views the asymmetry of the GABI as the result from the fact that North America had a higher pool of open-adapted taxa than South America. The opening of the isthmian region acted as an ecological filter favouring the dispersal of these open-adapted taxa to the detriment of the others. Hence, given the imbalance in open-adapted pool size between the two continents, more species would have been expected to disperse to the South than the other way around. The second one (B), formulated by Vrba (1992), proposes that the contraction of the tropical biome towards the equator triggered by the interglacial-glacial transition would have mechanically attracted tropical specialists southwards, irrespective of their continent of origin. Silhouettes are from phylopic, credits are available in Table S3. Note that the purpose of this figure is to illustrate our working hypotheses. Thus, we do not pretend these maps to accurately reflect inter-American biome dynamics between glacial and interglacial

periods. A more accurate picture of these dynamics is available on Figure S13.

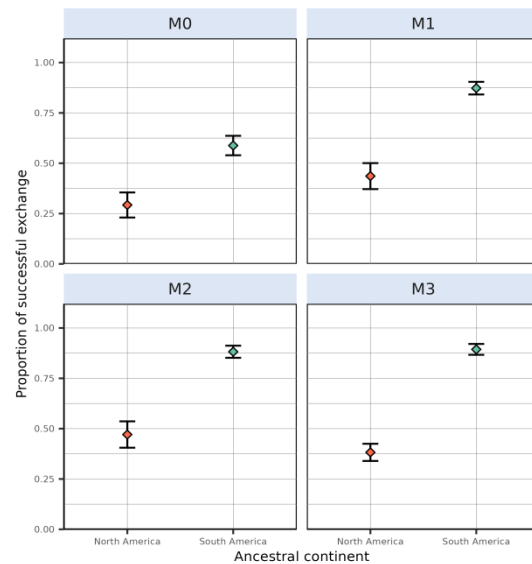


Figure 2: Proportion of simulations leading to a successful exchange to the other continent, computed as the ratio, given an initial continent and a modelling scenario, between the number of simulations leading to a transcontinental exchange and the total number of simulations resulting in at least one living species at the final state. From left to right are depicted the outcome of the zero-adaptationistfixed-niche model (*M0*, species climatic niche does not change through time—climate-specialists), full-adaptationistadaptive-niche model (*M1*, species climatic niche evolves following local changes in temperature and precipitations—climate-generalists) and alternative versions of *M1* where species ecological niche is only determined by precipitation (*M2*) or temperature (*M3*). Each individual plot shows the outcome of simulations starting in North (red) and South (cyan) America. Dots represent mean proportion estimates and error bars show their associated Binomial-derived 95% confidence interval.

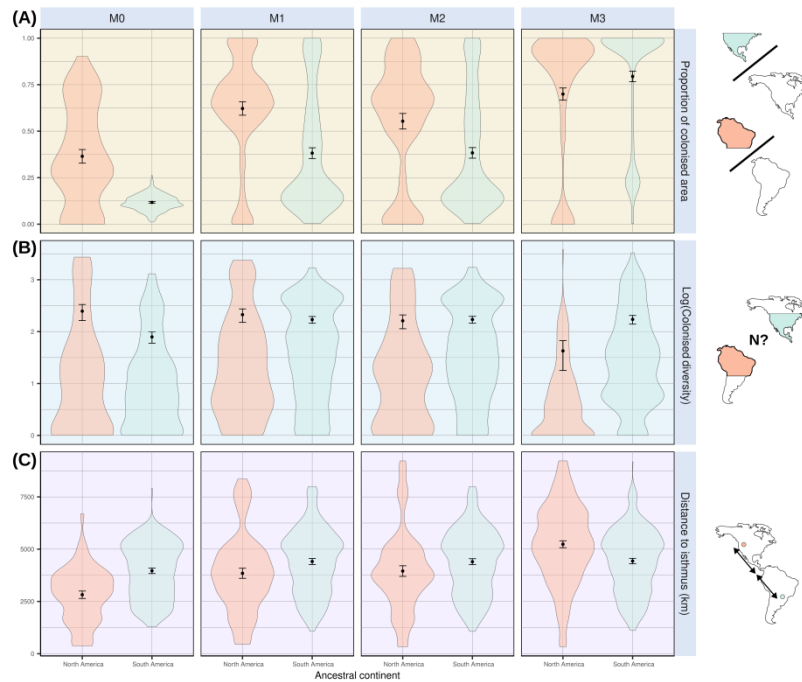


Figure 3: A few metrics summarising climate-mediated interchange given our four modelling schemes. derived These metrics were assessed for from successful simulations only, *i.e.*, simulations that resulted in a transcontinental faunal exchange. From left to right are shown the results of simulations generated under our four models (see Figure 2 for details). (A) Proportion of newly-colonised continent's newly-colonised area, quantified as the ratio between the number of pixels recording a presence in the landmass to colonise and the total number of pixels offered by this landmass. (B) Colonised diversity, expressed as the natural logarithm of the number of lineages within the colonised area. (C) Distances between the location where ancestral species originated simulation starting pixel and the isthmus of Panama, only for successful simulations — *i.e.*, successfully reaching the continent where it did not originate. Mean estimates and their associated Student-derived 95% confidence intervals s are displayed over their distribution.

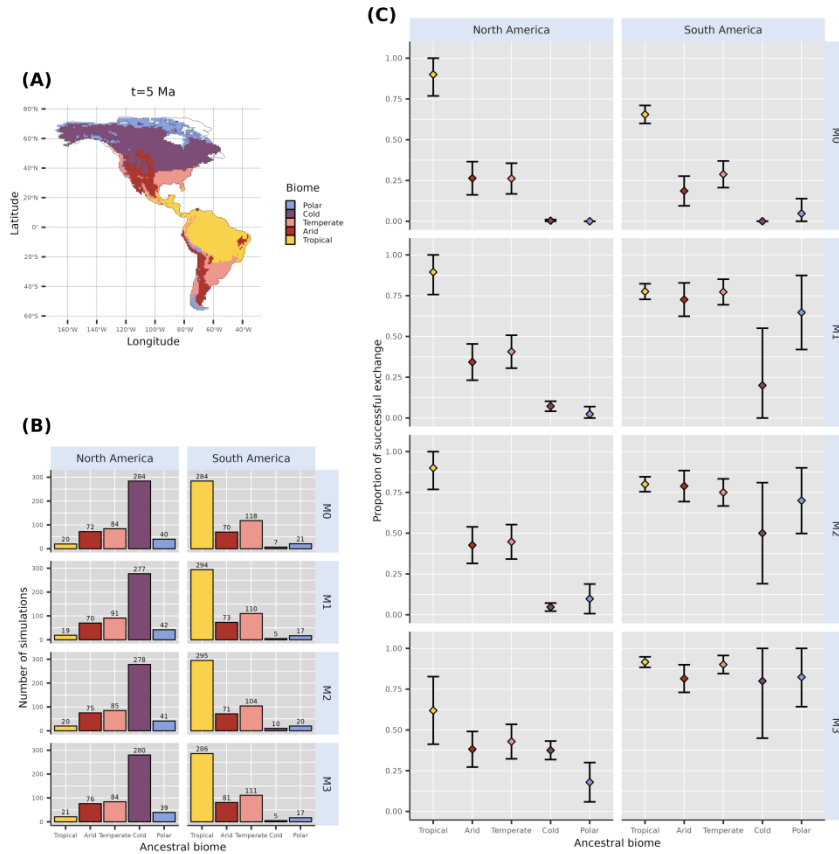


Figure 4: Initial biome where each ancestral species leading to successful simulations is recorded at the time of the onset of the simulations (5 Ma). **(A)** Global main climate zones based on Köppen-Geiger biome classification at $t = 5$ Ma. **(B)** Biome where the ancestral species of each of the 500 simulations originated, for each continent and each scenario. We removed simulations starting from a cell that could not be assigned a biome from our counts, hence the reason why some are not exactly summing to 500. **(C)** Proportion of successful simulations based on the biome where ancestral species originates, for each modelled scenario-ancestral continent combination. This proportion is expressed as the ratio between the number of successful simulations and the total number of simulations starting from a given biome. Coloured dots and error bars respectively show proportion estimates and their associated Binomial-derived 95% confidence intervals.

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Lucas Buffan
 Institut des Sciences de l'Évolution de Montpellier

May 29, 2026

Dear editorial board members of *Ecology Letters*,

We have received your decision letter concerning our manuscript (number 4175150), entitled **“Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange”**. We would like to thank you for considering our work, as well as for giving us the opportunity to revise it before a new submission. You may find enclosed a revised version of our manuscript and its *Supplementary materials*, which were modified following the comments and suggestions made by the reviewers.

Overall, the points raised by the Editor and the three referees were extremely relevant, and we believe they greatly improved our study. Noteworthy, as recommended by the Editor, we carried out additional batches of virtual experiment, allowing North American lineages to have broader thermal tolerance (i.e., increasing the range of the thermal breadth parameter associated to simulations starting in North America, ω_T). These supplemental sensitivity analyses strengthened our findings. In addition, we invited Dr. Philip B. Holden, the lead author of the palaeoclimate dataset that we use, to collaborate, in order to strengthen our reply to Reviewer 2's (highly relevant) worry about the dependency of our results to the palaeoclimatic reconstruction used. Dr. Holden was thus added as a co-author of the study.

We hope that the changes that we made to our manuscript will fit your expectations, as well as those of the three referees who kindly accepted to review our work. We wanted to stress that the substantial changes that our manuscript received after accounting for the issues raised over this review resulted in a 400-word inflation above the 5000-word limit set by *Ecology Letters*. Consequently, we had to move some paragraphs of the *Methods* section to supplementary materials. We hope that this will not alter the good understanding of this manuscript.

Below are the comments made by the Editor and the three referees (*italicised pink*), to which we respond with a normal font.

Editor's comments:

The manuscript has been reviewed by three reviewers, and their recommendations diverged. While all reviewers agreed that the manuscript is original and of high quality, praising the authors for using this modeling approach to address specific hypotheses, one reviewer raised concerns about the robustness of the results to the choice of climate models, the definition of boundaries between North and South America, and the delimitation of biomes. Another reviewer criticized the way the study is contextualized, contesting the claim that a climate-based hypothesis for the GABI asymmetry is widely accepted. During my own reading, I also identified some additional issues that could be addressed to make the main message clearer.

Thank you for this contextualisation.

Lineages with different origins are often compared under the same simulated scenarios (M0 to M3). Given these scenarios, one might expect the best fit to the observed patterns to arise from a model in which lineages originating in the South exhibit higher niche conservatism (M0), whereas those originating in the North show greater niche evolvability. Emphasizing the comparisons across models between lineages from different regions in the text and figures would help to demonstrate that, even in scenarios where southern lineages exhibit higher niche conservatism, the asymmetry does not emerge.

Given the empirical GABI pattern, one could indeed expect to retrieve smaller exchange intensities from fixed-niche simulations (M0) originating in South America than from evolving-niche (M1) ones originating in North

America. We agree that it would both improve the comprehension of the manuscript and strengthen our overall message to clearly state these assumptions while introducing our modelling schemes, and refer to them in the results section by explicitly emphasising relevant inter-model comparisons. Accordingly, we intended to implement these changes in the revised version of the manuscript. Thank you very much for this thoughtful remark.

In addition, the evolution of climatic niches is governed by a small number of parameters that are assumed to have the same range for lineages originating in North and South America. It would be interesting to explore the limits of these parameter ranges, for example, by expanding the range of sigma or the niche breadth for northern lineages under models M1–M3, to assess which parameter combinations might reproduce the observed pattern. This would help determine how asymmetric niche conservatism versus evolvability would need to be, and whether such asymmetry is biologically realistic, to account for the GABI pattern.

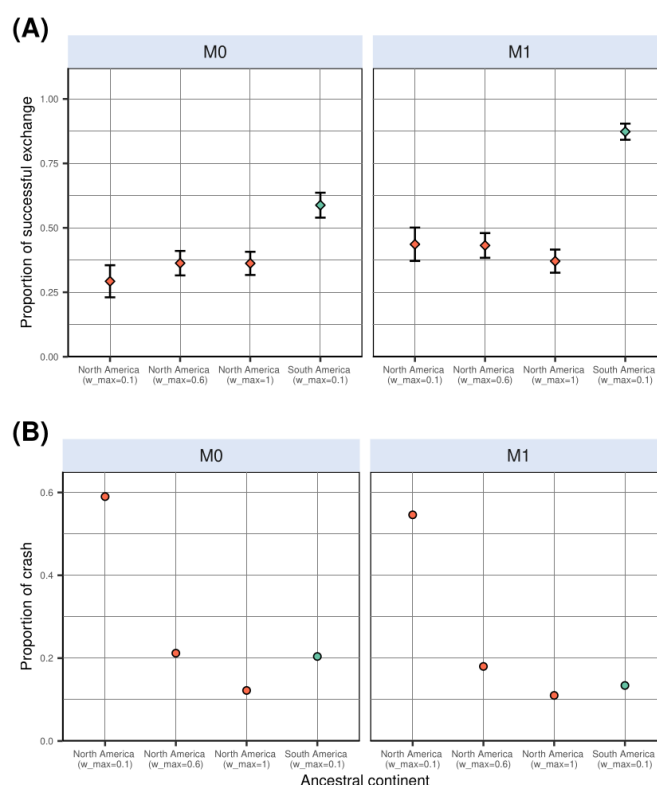
Thank you for this comment. We agree that testing for different niche breadths (ω_T or ω_P , resp. for temperature and precipitation) or evolvability (σ_e) ranges between simulations starting from North and South America could provide highly interesting results. Indeed, since North America's climate is more variable across time than South America's (as depicted on continent-averaged plots shown in Figure S1), the assumption that North American lineages would either have broader climatic tolerance, or could adapt faster to changing climatic conditions is highly relevant, and worth testing.

In the framework of our virtual experiments, we can expect that, irrespective of the model used to generate simulations (*i.e.*, allowing or not climatic niche to evolve), expanding ω of simulations originating in North America would increase the proportion of successful southwards exchange across the 500 replicates. The same assumption can be made on the basis of larger σ_e ranges for simulations starting in North America compared to those from South America. In addition, since climatic (and particularly temperature) temporal variability is more pronounced in North than in South America (see Figure S1), we can also expect that increasing ω_T range for simulations starting in North America would decrease the proportion of simulations crashing (*i.e.*, where the ancestral species goes extinct after a tiny number of iterations). Indeed, the proportion of simulations crashing was always 3-4 times higher when simulations started in North America than in South America (Table S2). Leveraging thermal constraints on species' ecology (*i.e.*, model M3) yielded almost no crash in the end (Table S2), suggesting that most crashing simulations were due to species' incapacity to cope with temperature change occurring within their home range.

We decided to focus on thermal niche breadth, and carried out additional experiments starting from North America, where we expanded the range of North American lineage's thermal breadth (ω_T) from [0.01, **0.1**] to [0.01, **0.6**] and [0.01, **1**]. We think that going beyond a 10-fold niche breadth difference between simulations originating in North vs. South America is not biologically realistic. We further compared the proportion of successful exchanges obtained with this experimental design to the one derived from our initial pipeline, starting from either continent, allowing (or not) climatic niche to evolve (M0 and M1).

These sensitivity analyses revealed that the inverse asymmetry in proportion of successful exchange that we characterised in our manuscript is robust to the aforementioned increase in thermal niche breadth for North American lineages (Supporting Figure 1A). Indeed, irrespective of the model under which simulations were generated (*i.e.*, M0 or M1), the proportion of successful North-to-South exchanges does not vary significantly when virtual species' thermal breadth is broadened by up to a factor 10 (Supporting Figure 1A). Also, in line with our expectations, increasing North American lineage thermal breadth drastically decreased (by a factor three) the proportion of simulation that crashed (Supporting Figure 1B). This explains the fact that, although the actual number of simulations leading to successful North-to-South exchange increased when we allowed North American lineages to have broader temperature tolerance, such an increase is also counterbalanced by the overall increase in the number of simulations leading to no exchange, and thus, that the overall proportion of successful exchange does not change when we allow North American lineages to have broader thermal niches.

We ensure to make the config files associated to this new set of analyses entirely available along with the rest of our data, on the same FigShare repository: <https://doi.org/10.6084/m9.figshare.31160704>



Supporting Figure 1. Comparative proportion of simulations resulting in successful inter-continental exchange (A) and of simulations that crashed (i.e., leading to no living representative at the end of the simulation) (B). Simulations were generated in a landscape based on the PALEO-PGEM series. `w\_max` indicates the upper bound of the thermal breadth parameter (ω_T) associated with the corresponding batch of simulations.

Considering the reviewers' comments and my own evaluation, I believe the manuscript would benefit from additional analyses and some revision before it can be considered for publication.

Reviewer #1:

The manuscript tests a clear set of macroecological hypotheses about the drivers of asymmetry in the Great American Biotic Interchange (GABI), using a well-designed modelling framework to evaluate whether spatio-temporal palaeoclimate dynamics can account for the observed pattern. The analyses are carefully executed, the study is well framed in the literature, and the results support the conclusion that climate alone is unlikely to explain the asymmetry, pointing to additional mechanisms that merit future work. Overall, the work is sound, clearly written, and well documented, and in my view, ready for publication. I enjoyed reading the manuscript. The methods are clear, and the modelling approach is well matched to the questions.

Thank you very much for this positive introduction. We are extremely delighted to read that you enjoyed our study.

Nevertheless, I would have liked a bit more discussion of limitations and future work in the main text (not only in supplementary material). For example: what would be the next modelling step? Which alternative hypotheses could explain the observed GABI asymmetry, beyond the "climate alone" expectation explored here? Are there mechanisms that would predict the opposite pattern to the natural tendency emerging from the climate-only simulations?

The reason why we moved the *Limitations & Perspectives* section to Supplementary Materials is that we were running out of characters in the main text and had to take a decision on how to reduce manuscript length to match the 5000-word limit imposed by *Ecology Letters*. We share your disappointment, and would have preferred to place it in the main text as well.

We agree that the perspectives that you raise could add interesting layers of discussion to the main text. However, we are afraid that the aforementioned manuscript length constraint prevents ourselves from doing so. That being said, in the third paragraph of the Limitations & Perspectives section – that we regretfully had to move to Supplementary Materials –, we intended to emphasise the (non-abiotic) factors that might have shaped the asymmetry of the GABI, and propose an avenue for further modelling studies. Also, in the main text, we intend to provide biologically-informed perspectives for the inverse asymmetry derived from our simulations, with examples of groups that, contrary to mammals, did not exhibit the classical patterns of higher success at colonising the other continent for Northern taxa. Although we are aware that it is a bit frustrating to constrain ourselves to such a tiny number of arguments in this section, we think that some of the points that you raised are intended to be covered. Hopefully this can be further expanded in follow-up studies!

Finally, the authors did an excellent job sharing the configs, landscapes, and code, and the literature coverage is strong.

Thank you!

One minor request: could the authors also provide the resistance values used or even better the resistance function when create_gen3sis_input function is run? (e.g., the value assigned to ocean / NA), ideally in the main text or a clearly referenced table?

That's a good point. Our choice of cost function was not innovative since it re-used the cost function designed for earlier studies (see screenshot below), making it twice as hard to disperse to a non-habitable (water) than to a habitable (land) cell (Skeels *et al.* 2023a, b). This function implied that we made the assumption that the cost of moving from a cell to another does not change across the simulation, and is not tied to any landscape change. This is indeed a strong assumption, and we agree that it should be mentioned in the *Methods* section of the main text. We therefore added it.

```
## Generate a gen3sis landscape -----
# Cost function (twice as high for water as for earth)
cost_function <- function(source, habitable_src, dest, habitable_dest) {
  if(!all(habitable_src, habitable_dest)) {
    return(2/1000)
  } else {
    return(1/1000)
  }
}
```

Minor comments

*Line 36. Add alone, since you can not exclude paleoclimate dynamics from the equation in case of other interacting mechanisms. Hence, we argue that spatio-temporal palaeoclimate dynamics ***ALONE*** cannot explain the asymmetry of the GABI, suggesting additional factors fashioned this macroecological enigma.*

Agreed, thank you.

Line 88. Liu et al. (2024) did not use process-explicit modelling for the biological component; that part was correlative. The process-based component was applied to landscape evolution only.

That's correct. We cited it because it is recent and is not a gen3sis case study (contrary to the other ones cited alongside), but indeed it would have made more sense to cite a paper using another MEEM rather than a process-explicit model of landscape evolution. We removed the citation.

Lines 102–105. Nice job keeping the continent names consistent and clearly defined.

Thanks!

M0 and M1 labels. Consider renaming M1 and M0. The labels “generalist” and “specialist” may not reflect the adaptation mechanisms you tested. They can also be read as referring mainly to niche breadth. More neutral labels (e.g., “adaptive niche model” vs. “fixed niche model” or something related with niche breath – see next comments) may better match the design.

This is totally accurate. It actually makes more sense to read the generalist vs. specialist dichotomy through the lens of niche breadth – a parameter that was held constant across simulations – rather than through the evolvability of the optimum parameter. We did not consider it, so thank you for stressing this. We modified the manuscript accordingly, adapting the phrasing when necessary.

Consider introducing tropical niche conservatism earlier (rather than first at ~line 357). It is a key concept that connects directly to your M1/M0 framing and overall study design.

You are right that tropical niche conservatism is quite a central concept that would deserve to be mentioned before the discussion. Consequently, we introduced it after the *Methods* section paragraph introducing our modelling schemes. Thank you for the suggestion.

Reviewer #2:

The analysis wants to test if paleoclimate alone can explain the mammal South America (SA)-North America (NA) asymmetry, e.g., extant mammal faunas have more NA lineages in both SA and NA. This pattern was formed following the connection of SA and NA in an event known as GABI (the Great American Biotic Interchange). Three distinct and well-thought hypotheses are to be tested (i) a NA ancestor would be more successful at reaching the other continent than viceversa, (ii) a NA ancestor migration would result in a more diverse and spatially spread radiation in the newly-colonised continent, and (iii) successful interchange would be more frequently achieved among biome generalists, extratropical specialists rather than tropical specialists. The analysis uses a paleoclimatic model (PALEO-PGEM) and an eco-evolution simulator (gen3sis). The modeling exercise is well done, following a logical pathway reaching the conclusion that paleoclimate alone cannot explain the asymmetry and actually, the opposite pattern would have been expected (more migrations from SA to NA). This is a very interesting result that will open a new venue for research on GABI, one of the largest biogeographic experiments of the Cenozoic. The manuscript is well written, and the illustrations are appropriate; it would be a good fit for the journal.

Thank you very much for these opening words. We are happy to read that you overall appreciated our work.

However, I see major problems:

1. Modeling climate is a difficult task. Consequently, Global Circulation Models have major prediction discrepancies for predicting future and past climates, especially precipitation in both the tropical band and in the center of large continents. The study used the PALEO-PGEM model, but there are many others (e.g., bcc-csm1-1, CNRM-CM5, COSMOS-ASO, GISS-E2-R_p150, GISS-E2-R_p151, IPSL-CM5A-LR, FGOALS-g2, MIROC-ESM, MPI-ESM-P_p1, MRI-CGCM3, CCSM4, CSIRO-Mk3-6-0, CSIRO-Mk3L-1-2). Some of them predict that Panama was a savanna during the LGM, while others indicate that there was a rainforest. How dependent are the results on the chosen climatic model? That is a question that must be quantified for this study to be helpful. I think the study needs to select another climate model that predicts contrasting differences with PALEO-PGEM model, to assess if the pattern found in the study holds, or it is highly dependent on the climatic model.

Thank you for this very thoughtful comment. Our simulations are indeed exclusively tied to the assumptions and the outputs of a single GCM, PLASIM-GENIE (Holden *et al.* 2016), statistically emulated across the last 5 Myrs with a 1000-year interval by the PALEO-PGEM emulator (Holden *et al.* 2019), leading to the PALEO-PGEM series (Barreto *et al.* 2023). As you stress, many different GCMs exist, differing in their assumptions, the data they rely on and, in turn, in their predictions. For that reason, testing the sensitivity of our results to the palaeoclimate reconstruction used would be highly relevant and, to our knowledge, has never been done in eco-evolutionary studies using gen3sis.

One constraint arises as, despite the existence of a vast number of GCMs, most of them have a limited number of scenarios (mainly LGM, mid-Holocene and Present). Yet, our eco-evolutionary simulation framework requires palaeoclimate layers temporally interpolated between each others with a narrow resolution

(10 ky-interval) throughout the last 5 million years. Such a resolution is typically available for the PLASIM-GENIE model (via the PALEO-PGEM series), but not for many other GCMs often used in macroecology, including those you listed, making it impossible for us to use the outputs of these GCMs in our pipeline.

To our knowledge, PALEO-PGEM-Series is the only palaeoclimate dataset that has both sub-orbital temporal resolution and extends back to 5 Ma, as needed for our study. One alternative dataset that comes close to our needs is the 3 Ma transient coupled simulation with CESM1.2 (Yun *et al.* 2023), made tractable by splicing together 42 simulations that were run in parallel. Unfortunately, this does not encompass the Pliocene and would thus only allow to test for biotic exchange occurring during the Quaternary, narrowing the scope of our work. In contrast, the Oscillayers dataset (Gamisch 2019) does extend back to 5.4 Ma, but the approach of pattern scaling LGM anomalies does not represent orbital variability – the dominant driver of climate variability at 10 ka. Furthermore, it unrealistically assumes that Pliocene warming has the same spatial pattern as LGM cooling. Similar concerns were expressed by (Brown *et al.* 2020). Although Oscillayers superficially meets our requirements in terms of spatio-temporal coverage and resolution, this dataset is not fit for our purpose, and in our view should not be used in the Pliocene for any application. Consequently, at time of writing, the sensitivity of our pipeline to the palaeoclimate reconstruction used cannot be evaluated, which represents a strong limitation to our work.

We incorporated these elements in an additional paragraph of our *Limitations and Perspectives* section.

2. Figure 1 shows Central America and western Amazonia/SA to be savanna/desert, while only a small pocket is rainforest in eastern Amazonia. I wonder where this biome interpretation came from. I downloaded the Paleo-Pgem dataset from the Barreto article and plotted precipitation for the LGM and the pattern seems quite different to the biomes presented in figure 1 (see attached figure). I searched into the supplementary material, the actual MAT and MAP maps for each time slice that were used, but I could not find them. It would be useful if a few time layers are shown (e.g., today, the LGM, the last Interglacial, 1 My ago, 2.6 My, 3 and 5 My).

Figure 1 directly draws on the palaeoenvironmental interpretations made by Webb (1991), that were most likely extrapolated for artistic/visual purpose (see below a screenshot of the figure in question, along with its caption). The purpose of Figure 1 was more to set the scene and present the hypotheses that are being tested in the manuscript rather than to provide an accurate picture of landscape change throughout the study period. Raw first-order maps approximating biome temporal dynamics with illustrative purpose can be found in the literature. A good example is Figure 1 of Meseguer & Condamine (2020), depicting the evolution of the extent of the tropical belt through the Cenozoic to show how it may have influenced the establishment of the Latitudinal Diversity Gradient. To avoid further confusion, we extended Figure 1's caption to clarify that these maps do not pretend to depict the most accurate picture of biome dynamics in the Americas between glacial and interglacial phases, and that their purpose is purely illustrative.

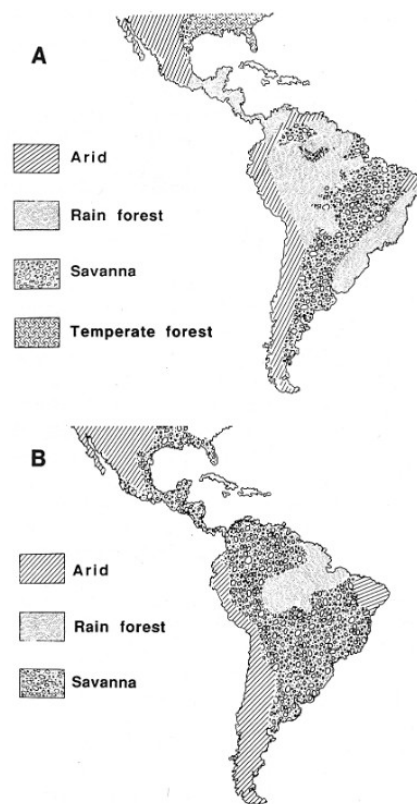
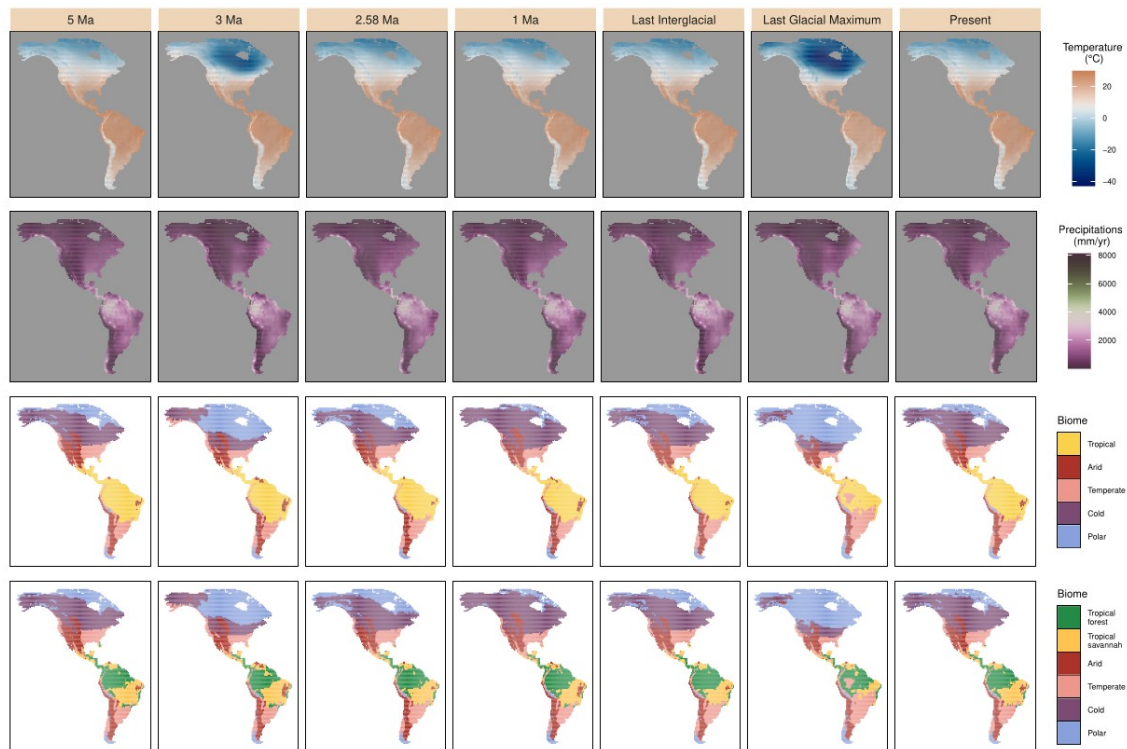


FIGURE 3. Approximate distribution of four major landscapes in the American tropics during two phases of the Late Cenozoic. *A*, interglacial phase dominated by equatorial rainforest including the critical isthmian link. At this time Amazonian biota conquered Central America and southern Mexico. *B*, glacial-phase; savanna and other open landscapes predominated. At this time, the large savanna reservoirs in North and Central America transmitted far more interchange taxa than the smaller areas in temperate South America could retain.

In addition, we did not make the palaeoclimatic layers that we used available along with our publication to avoid useless data storage, since – as indicated in the Methods section – these were directly downloaded from the supplementary materials of Barreto et al. (2023) without any subsequent change (the files in question were called `bio1\_mean.txt` and `bio12\_mean.txt`, stored in the `biovars\_mean.zip` folder), as you did to produce your figure. Biome classification relied on the monthly average temperature and precipitation available in the `monthly\_mean.zip` folder. That being said, we agree that inserting some palaeoclimate variable snapshots at precise time steps could help better visualise and apprehend the data our entire study relied on – *i.e.*, the palaeoclimate layers. Consequently, we inserted a panel in supplementary materials, depicting Bio1 and Bio12 at precise snapshots back in time, along with the corresponding biome reconstructions (see figure below). This panel clearly contrasts with the caricatural fragmentation that we drew on Figure 1. During glacial maxima (*e.g.*, LGM), the fragmentation of the closed tropical biomes (tropical rainforest and monsoon) is way less pronounced than on our representation, but we believe that leaving Figure 1 as it is should not be a problem as long as we clearly state that its aim is purely illustrative (which we intended to do by accounting for your comment). Last, it should also be noted that palaeoclimate reconstructions at the beginning of the 1990s, when the climate-driven asymmetry hypotheses were formulated, were not as advanced as what they are now, explaining potential mismatches between 1990s reconstructions and those that we obtained.



Snapshots of the two bioclimatic variables used in this study (Annual Temperature and precipitations, respectively called Bio1 and Bio12 in the climate model) from PALEO-PGEM (Holden et al. 2019), together with the corresponding biome reconstructions that we carried out following Galván et al. (2023) (using broad climate classes), at key time steps. In the biome reconstruction that we carried out for the bottom row, the "Tropical" class was subdivided into "Tropical forest", incorporating closed Rainforest and Moist forest (Monsoon), and "Tropical savannah", associated with a more open landscape. Last Interglacial corresponds to $t=120\text{ka}$ and Last Glacial Maximum to $t=21\text{ka}$.

3. How were the main five biomes determined? (the climatic boundaries that were chosen for each biome must be explicit).

In the *Palaeoclimate data* section, we explained that: “we reclassified monthly palaeoclimate layers for our starting time ($t=5\text{ My ago}$) according to Köppen-Geiger broad climate types (Beck et al. 2018; Galván et al. 2023). This classification divides the world into five broad climatic regions: Polar, Cold, Temperate, Arid and Tropical, following the criteria proposed by Köppen (1900)”.

4. I could not find a single mammal taxon anywhere in the metadata. There should be a list with the name of the taxa, time of origination, time of extinction, site of origination (lat, long), biome of origin, and additional biomes where it occurred along its history. Very hard to assess the results if none of the raw data is shown.

Sorry for the confusion, but this work exclusively relied on simulated data, not on true mammal taxa. Apart from the palaeoclimate estimates, our pipeline did not incorporate any empirical datum. Our aim was to assess the extent to which a simple simulation framework relying on palaeoclimate could capture a sufficient amount of complexity to explain the asymmetry behind the GABI. As the simulations failed to reproduce the asymmetry, we did not intend to match the simulated diversity pattern with actual data (either of fossil or extant taxa).

Furthermore, it would be useful if some lineages are discussed. For example, I have always puzzled over the horses, camels, and rhinoceros found in the lower Miocene tropical rainforests of Panama. All three lineages would be classified as occupying open biomes, and yet they inhabited the rainforest of Panama for millions of years. Perhaps, they reached newly-formed forest still lacking South American mammals, thus being able to establish even though they moved into a new biome

Since our study is entirely simulation-based, assigning empirically-relevant biological groups to virtual lineages to enhance such a discussion may not be straightforward. That being said, the point that you raise is very interesting. Indeed, as you indicate, particularly in the case of land mammals, some Late Miocene faunal

assemblages from Central America (e.g., Gracias Fm in Honduras and El Salvador) were almost exclusively composed of genera also found in temperate North America. These include, among others, rhinocerotids (*Teleoceras*), equiids (*Hipparion*, *Pliohippus*), camelids (*Procamelus*, *Protolabis*), bovids (*Bison*) or gomphotheriids (*Rhynchotherium*, *Cuvieronius*) (Webb & Perrigo 1984) (note that almost none of the aforementioned taxa made it to South America). Despite uncertain palaeoenvironmental context for some localities, it cannot be excluded that these assemblages deposited in a tropical forested environment, which contrasts with the conspicuousness of the taxa in question in clearly-established temperate assemblages of North America. Also noteworthy, these assemblages had a poor level of endemism, meaning that most of the taxa that they revealed were also found elsewhere (e.g., temperate North America).

The presence of these taxa in these (likely) tropical forest assemblages may be interpreted under the lens of the enhanced facility for North American non-tropical groups to colonise tropical areas, as evidenced by Smith et al. (2012), who characterised an asymmetric niche conservatism between North and South American land vertebrate families. Following their result, in the framework of the Americas, colonising tropical areas for non-tropical taxa would be easier than the other way around.

Due to space limitation, we could not expand ourselves as much as we would have liked to in this respect, but we believe that this aspect is already approached in the *Discussion* paragraph in which we analyse our findings under the prism of tropical niche conservatism.

5. The NA vs SA delimitation was set in western Panama. That is a strange boundary that does not correspond to the geological boundary between the Caribbean plate (Panama microplate) and the South American plate. The actual geological boundary between CA and SA is along the Uramite suture in the western cordillera of Colombia (e.g., Montes et al 2015, Science)

As indicated in the main text, the delimitation between North and South America was placed at random somewhere in Panama. As you raise, this choice does not reflect the geology of the continents, instead placing such a separation along the Uramite suture. Our choice reflected some assumptions of connectivity between the landmasses prior to the onset of our simulations (5 Ma). Indeed, placing the delimitation along the Uramite suture would suggest that the closure of the Central American Seaway (CAS), in the middle Miocene (Montes *et al.* 2015), would have marked the full closure of the isthmus of Panama. Despite the fact that the closure of the CAS most likely interrupted deep-water circulation between the Caribbean sea and the Pacific ocean, there are compelling evidence suggesting that shallow water flow persisted through Panama at least until the Pliocene (Jaramillo 2018; O'Dea *et al.* 2016 and literature therein). We assume this water flow persistence following the closure of the CAS to have acted as a barrier against most terrestrial exchanges between the continents (though some exceptions exist, see Woodburne 2010). Thus, at the beginning of our simulations (*i.e.*, virtually following the closure of the isthmus), it is relevant to consider that South American lineages could reach a piece of Panama. That being said, moving the inter-continental delimitation a few cells southwards should not impact the overall simulations' outcome.

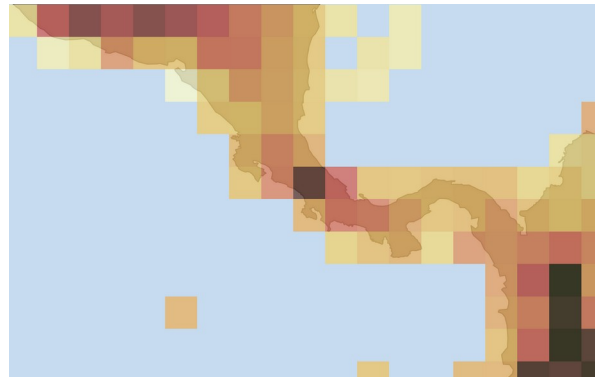
6. It is possible that the Andes acted as a conduit for temperate taxa to cross tropical latitudes in both directions. However, a grid of 1 degree would fail to capture the granularity of the Andes climates, and probably the MAT would be lower than the actual values (the Andes grid would contain areas from the lowlands, increasing the MAT for the grid). Perhaps a grid approach is not appropriate for the Andes, and it would be better to use actual shapes that follow the Andes contour lines (e.g., 500-2000m, 2000-3800m, >3800), which would move down during glacial times. Same occur with the shallow bays all along Central America, as all became land during glacial times. That would increase land area by 45% in Panama alone.

Considering that the Andes acted as a corridor for temperate North American taxa to disperse southwards is full of sense. It could explain the successful colonisation of taxa such as camelids in South America – though the fossil record suggests that, in the Pleistocene, camelids could also be found in the plains, as attested by the Eastern Brazil occurrences of *Palaeolama*. One could also think about the astonishingly diverse radiation of sigmodontine rodents, in which the Andes were shown to have played a key role (Vallejos-Garrido *et al.* 2023).

As you notice, part of the palaeotopographic information is reflected by the palaeoclimate layers. Although a significant amount of local altitude microclimate information (likely important for the dispersal of

several taxa) is probably lost due to the usage of a $1 \times 1^\circ$ grid, we believe that our approach does not erase all the mountain signal, and that it would be hard to gain more accuracy in that respect.

Next, regarding the changes in the extent of coastlines in shallow bays (e.g., Panama) between glacial maxima and interglacials, your remark is totally relevant. Indeed, this must have significantly changed the amount available area, particularly in the zone of connection between the North and South American landmasses. However, we think that such variations lie within the boundaries of our PALEO-PGEM-derived landscape (not changing across simulations, which is indeed a limitation that we intend briefly mention in the *Limitations and Perspectives* section), as depicted on the plot below.



Spatial extent of PALEO-PGEM for Central America. Each pixel is a $1 \times 1^\circ$ cell, filled with an annual temperature (the value range is not important here, all what we wanted to show is that PALEO-PGEM's grid cells span a large area in Panama). Note that the isolated pixels, representing islands, were excluded from our actual simulation landscape.

This is a good piece of work that is worth publishing. However, it requires running an additional experiment (at least one more GCM) and providing the taxa that were used in the analysis.

Thank you for these very useful comments. We believe that they greatly improved the study.

Reviewer #3:

I am not sure what is the right recommendation for the manuscript but I lean towards a major revision. The main issue lies in the overall framing. The manuscript claims that it is the common expectation that GABI can be explained by climate alone and that it therefore is a surprise that they cannot do it. That seems to be a very one-sided reading. There are ample studies arguing that the climate model is not the most likely. Some are cited but dismissed without good reasoning. Another very cool one is still in preprint only but very central in this regard (<https://www.biorxiv.org/>) If the paper was rewritten it would be interesting to show with simulations as these that a model lacking good theoretical basis also fails with simulations. Climate is still a commonly claimed driver and such a paper would clearly be publishable and I can easily see myself citing it, but I am not sure if such a paper would be strong enough for a journal like ecology letters? In its current form I cannot recommend publication anywhere. In a rewritten form I would recommend publication somewhere which may or may not be this journal. In addition to this major issue there are a few minor things I will address below on a line by line basis

We apologise if our manuscript gave the impression to be built upon the fact that climate-driven asymmetry of the GABI was “*the common expectation*”. We acknowledge that alternative hypotheses have been proposed – that, as you noticed, we intend to report in the second paragraph of the introduction –, and that constraining the complexity of such a major biogeographic event to climatic changes is probably inaccurate. That being said, the hypotheses invoking palaeoclimate as the only driver behind such an asymmetry have been formulated for decades (Vrba 1992; Webb 1991), have maintained in the field since then (Bacon *et al.* 2016; Woodburne 2010), and were supported by recent findings (Freitas-Oliveira *et al.* 2024a, 2025 and literature therein). They stood out

as an alternative to the long-standing view that the asymmetry of the GABI was the result from biotic interactions (Simpson 1950), either competition or predation, that were contrastingly supported in the recent literature, championed by some studies (Domingo et al. 2020; Faurby & Svenning 2016; Pino et al. 2025) but invalidated by others (Freitas-Oliveira *et al.* 2024b; Pino *et al.* 2022; Prevosti *et al.* 2024; Tarquini *et al.* 2022). Consequently, we do not share the view that the abiotic-only hypotheses are “*lacking good theoretical basis*”, and still think that such hypotheses are worth testing.

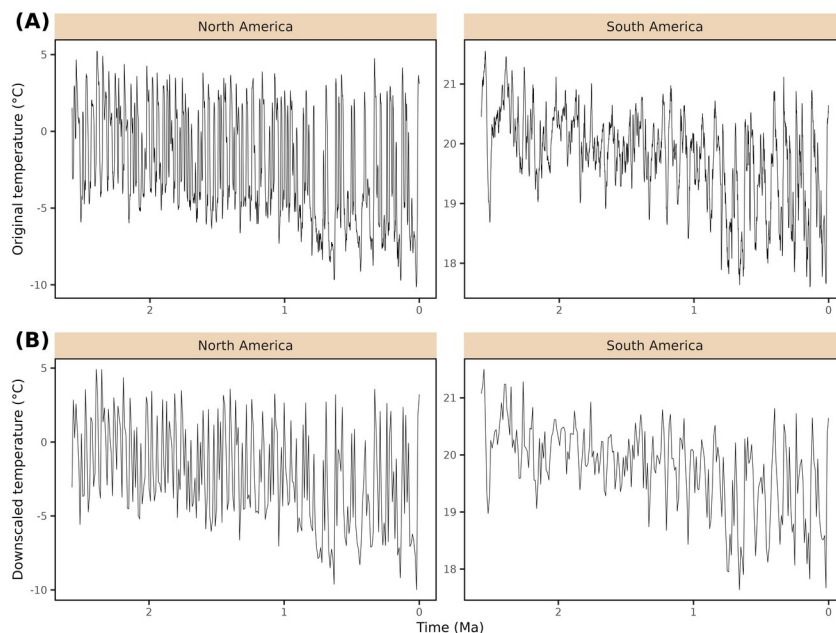
We never meant to “*dismiss without good reasoning*” the wealth of hypotheses relating the asymmetry of the GABI to non-climatic factors. Given the complexity of the studied pattern and the immense variability that a simulation framework like gen3sis is likely to generate, we instead think that testing alternative hypotheses would deserve to be the topic of an entire study in itself (or even several ones), and that going beyond testing the abiotic-only hypotheses would have been infeasible in a single paper – especially in a relatively short format, as the one imposed by *Ecology Letters*.

As a result, we argue that our study should be more seen as a first attempt to model the GABI with eco-evolutionary simulations, testing long-standing hypotheses, and that can be used as a building-block for improved processed-based modelling of the GABI, rather than an attempt to build the “*most likely*” model of the GABI. From a broader perspective, the idea that climate change impacts the temporal dynamics of life on Earth is deeply rooted in the fields of biogeography and palaeontology (Erwin 2009; Malanoski *et al.* 2024; Nogués-Bravo *et al.* 2018), adding some relevance to testing hypotheses relating biodiversity and climate dynamics, which is what is intended in this study.

Lines 109-112.

I think that some argument or SI results is needed for why the change in window does not change the amount of climatic oscillation in the Pleistocene. If the timewindow gets too big , the interglacials may disappear. I do not think this should be a major problem for 10,000 years but I think some argumentation and/or SI plot is needed .

That’s a good point, thank you. We added the plot attached below to the Supplementary Materials. You may see that, as you expected, upscaling temperature reconstructions from a 1000-year to a 10,000-year interval had little effect on the trend of the variable across the Pleistocene; we still recover the oscillations. We mentioned it in the main text.



Effect of climatic variable downscaling. Continent-averaged temperature estimates were produced for both North and South America, with the original time step of the climate emulator (1 ky, A) or after downscaling to a 10 ky time step (B).

Figure 3.

Somewhat confusing explanation in the legend and the pictures did not really help me either. I think the figure or legend should be remade.

Thank you for pointing this out. We modified Figure 3's caption to make it more understandable.

Lines 315-317.

I think that a few words may be needed on the claim that hummingbirds are a South American group. Based on current species this would clearly be the claim but ancestral area reconstruction is known to be very poor without incorporating fossils, see e.g. <https://doi.org/10.1098/rspb>, and the oldest fossils of humming birds are European. It seems highly plausible that hummingbirds died out in North America long before the GABI and that it makes sense to consider it South American in this context but with the very scarce fossil record of birds and the geography of the few known fossils in the family I think caution is needed in word choice.

Thank you for this thoughtful remark. Your point is highly relevant; the European fossil hummingbirds of Oligocene age suggest that the deep historical biogeography of the group is clearly not exclusively tied to South America. These fossils are regarded as stem Trochilidae (Mayr 2004), and phylogenetic evidence have suggested that the crown group would have radiated in the Miocene of South America, and that all the extant hummingbird lineages that colonised North America (e.g., bee, coquette) descent from a South American lineage (McGuire *et al.* 2014), although we share your concern about carrying out historical biogeographic reconstructions ignoring the (or in the absence of) fossil record. A similar mismatch between the distribution of the crown radiation and fossil occurrences of stem representatives has been described among other purely Neotropical extant radiations, such as Tyrannida (Harvey *et al.* 2020; Riamon *et al.* 2020). Consequently, we modified our sentence “such as hummingbirds” → “such as **crown** hummingbirds”.

Line 346-348.

It is hard to reliably say anything for sure about habitat preference for extinct groups especially given fossilization biases but at least for proboscids it seems odd to me to call them an open area group. Living elephants do perfectly find in forested habitats after all.

This is correct. After checking the literature (e.g., Asevedo *et al.* 2012; Sánchez *et al.* 2004), the trophic preferences of the two proboscid genera that made it to South America (especially *Notiomastodon*) could clearly match a forested environment. We removed the mention of the group in the sentence. Thanks for the suggestion.

If we can answer any further questions, or be of any assistance to you whatsoever, please feel free to contact us. Thank you for considering our revised manuscript.
Yours sincerely,

Lucas Buffan, on behalf of my co-authors



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Lucas Buffan
Institut des Sciences de l'Évolution de Montpellier

May 25th, 2026

Dear Editor-in-Chief and Editorial Board members of *Ecology Letters*,

Please find enclosed our revised manuscript entitled “***Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange***”, which we are pleased to submit for a second time for consideration as a letter in *Ecology Letters*.

Our manuscript received substantial changes that we intend to thoroughly describe in the response letter submitted alongside the revised version of our study.

Feel free to contact me if you need any further information.
Yours sincerely,

Lucas Buffan, on behalf of my co-authors

A handwritten signature in blue ink, appearing to be 'L. Buffan', written over a horizontal line.

Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

Lucas Buffan^{1*}, Fabien L. Condamine¹, Philip B. Holden², Sara Varela³

¹ Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France

² Environment, Earth and Ecosystem Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK

³ Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

* Corresponding author: lucas.l.buffan@gmail.com

Statement of authorship

S.V. and L.B. designed the project. P.B.H. gave inputs on palaeoclimate data usage. L.B. ran the simulations, performed the analyses, produced the figures and wrote the manuscript with inputs from all co-authors.

Data availability statement

R scripts for generating, analysing and visualising the data are available at <https://github.com/Buffer3369/GABI-gen3sis.git>. Simulation landscapes and config files are available at <https://doi.org/10.6084/m9.figshare.31160704>.

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